STUDENT PRACTICE EDITION - FULL LAYOUT DRAFT

MOTION IN A STRAIGHT LINE

Transfer Question Book

Work one set at a time. Attempt first; H1 -> H2 -> H3 only when needed.

START HERE

SET 1 / 8

Average speed & average acceleration

Do not browse the whole book. Finish the current set, then move forward.

STUDENT EYE-PATH

QUESTION -> WORK HERE -> STOP LINE -> H1 NOTICE -> H2 MODEL -> H3 START

H1-H3 may include numbers and intermediate equations. Each level is deliberately more concrete.

HOW TO USE THIS BOOK

1. Read only the current question.
2. Use the blank work area before looking down.
3. If stuck, read H1 only. Continue working.
4. Use H2, then H3 only if you still need help.
5. After attempting, open Check Your Method and return.

8 STUDY SETS

SET 1 / 8

Average speed & average acceleration

Work first. Hints stay below the work area.

Q1 D3 M02B

EX-M2B-01

QUESTION

Half the journey is at 6 m/s. The remaining half is covered in two equal time intervals at 9 m/s and 15 m/s. Find the average speed.

JEE MAIN 2024 - 9 APR MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	The second half uses equal times at 9 and 15 m/s, so its speed for that half is (9 + 15)/2 = 12 m/s.
H2 MODEL	Let total journey = 2L. First-half time = L/6; second-half time = L/12.
H3 START	Average speed = 2L / (L/6 + L/12). Now simplify.

Q2 D1 M02B

EX-M2B-02

QUESTION

A person travels distance x at velocity v_1 and then another equal distance x at velocity v_2 in the same direction. Relate the average velocity v to v_1 and v_2.

JEE MAIN 2023 - 10 APR EVENING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Both legs have the same distance x. Their times are x/v_1 and x/v_2.
H2 MODEL	Total distance = 2x; total time = $x/v_1 + x/v_2$.
H3 START	$v_{avg} = 2x / (x/v_1 + x/v_2)$. Cancel x and simplify.

Q3 D2 M02B

EX-M2B-03

QUESTION

An object covers three equal consecutive distances at speeds v_1, v_2 and v_3. Find the average speed.

JEE MAIN 2023 - 1 FEB MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — Let each equal distance be L. The three travel times are L/v_1, L/v_2 and L/v_3.

H2 MODEL — Total distance = 3L; total time = $L(1/v_1 + 1/v_2 + 1/v_3)$.

H3 START — $v_{avg} = 3L / [L(1/v_1 + 1/v_2 + 1/v_3)]$. Cancel L.

Q4 D1 M02B

EX-M2B-04

QUESTION

A vehicle travels 4 km at 3 km/h and another 4 km at 5 km/h. Find the average speed.

JEE MAIN 2023 - 30 JAN EVENING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — The two distances are equal: 4 km each. Times are 4/3 h and 4/5 h.

H2 MODEL — Total time = 4/3 + 4/5 = 32/15 h; total distance = 8 km.

H3 START — Average speed = 8 / (32/15). Now simplify.

Q5 D1 M02B

EX-M2B-05

QUESTION

A car travels distance x at speed v_1 and then the same distance x at speed v_2. Find average speed.

JEE MAIN 2023 - 25 JAN MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — The two legs are equal distances x. Their times are x/v_1 and x/v_2.

H2 MODEL — Total distance = 2x; total time = $x/v_1 + x/v_2$.

H3 START — Average speed = $2x/(x/v_1 + x/v_2) = 2v_1v_2/(v_1+v_2)$.

Q6 D2 M02B

EX-M2B-06

QUESTION

A person moves distance x at 5 m/s and the next 3x/2 at speed v_2. The overall average velocity is 50/7 m/s. Find v_2.

JEE MAIN 2025 - 2 APR MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — The distances are x and 3x/2, so total displacement = 5x/2.

H2 MODEL — Total time = $x/5 + 3x/(2v_2)$.

H3 START — Set $(5x/2)/(x/5 + 3x/(2v_2)) = 50/7$ and solve for v_2.

Q7 D3 M02B

EX-M2B-07

QUESTION

A horse rider covers half the distance at 5 m/s. The remaining half is covered at 10 m/s and 15 m/s for equal times. The mean speed is x/7 m/s. Find x.

JEE MAIN 2023 - 30 JAN MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — For the second half, equal times at 10 and 15 m/s give an equal-time average of 12.5 m/s.

H2 MODEL — Let total distance be D. First-half time = D/10; second-half time = (D/2)/12.5 = D/25.

H3 START — Average = D/(D/10 + D/25) = 50/7 m/s, so compare with x/7.

Q8 D2 M02B

EX-M2B-08

QUESTION

A car covers three equal one-third portions at 11, 22 and 33 m/s. Find average velocity.

JEE MAIN 2022 - 28 JUN EVENING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — The three portions are equal distances. Use one common distance L.

H2 MODEL — Total time = L/11 + L/22 + L/33; total distance = 3L.

H3 START — Average = 3/(1/11 + 1/22 + 1/33).

Q9 D3 M03A

EX-M3A-01

QUESTION

A tennis ball is dropped from 9.8 m and rebounds to 5.0 m. Contact with the floor lasts 0.2 s. Take $g=10\ m/s^2$. Find average acceleration during contact.

JEE MAIN 2023 - 29 JAN MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE

Choose upward positive. Just before impact the ball is moving downward; just after rebound it is moving upward.

H2 MODEL

From the given heights, speeds are 14 m/s before impact and 10 m/s after rebound, so v_{before} = -14 and v_{after} = +10 m/s.

H3 START

Average acceleration = Delta v/Delta t = [10 - (-14)]/0.2.

Q1 D1 M4A-C5 **EX-M4A-01**

QUESTION

A particle moves in +x with u = 10 m/s and constant a = 2 m/s^2. How long to reach 60 m/s?

TRANSFER QUESTION

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Distance is irrelevant. The known quantities are u=10 m/s, v=60 m/s and a=2 m/s^2.
H2 MODEL	Use the constant-acceleration relation v=u+at.
H3 START	Substitute 60 = 10 + 2t and solve for t.

Q2 D2 M4B-C8 **EX-M4B-01**

QUESTION

A bus moving at 72 km/h is brought to rest uniformly in 4 s. Find the braking distance.

TRANSFER QUESTION

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	72 km/h = 20 m/s. "Brought to rest" gives v = 0.
H2 MODEL	One uniform-braking interval. First find a from v = u + at, then find s.
H3 START	Convert 72 km/h = 20 m/s. With v = 0 and t = 4 s, 0 = 20 + 4a -> a = -5 m/s^2. Then s = ut + 1/2at^2 = 20(4) + 1/2(-5)(16) = 80 - 40 = 40 m.

Q3 D3 M4B-C6

EX-M4B-03

QUESTION

A body starts from rest with constant acceleration. It covers S_1 in the first (p-1) s and S_2 in the first p s. In what time is displacement S_1+S_2 covered?

TRANSFER QUESTION

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

From rest with constant a, cumulative displacement has the form Kt^2.

H2
MODEL

$S_1 = K(p-1)^2$ and $S_2 = Kp^2$. Their sum must equal KT^2.

H3
START

SET UP THE EQUATION $T^2 = (p-1)^2 + p^2 = 2p^2 - 2p + 1$. Take the positive square root for time.

Q4 D2 M4B-C2

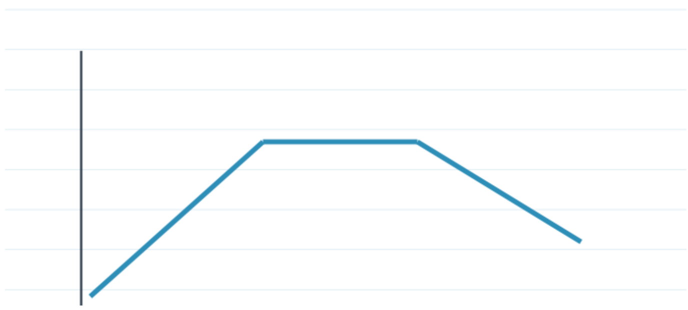
EX-M4B-04

QUESTION

A particle starts at the origin and moves along +x with the shown v-t graph. Find its position at t = 5 s.

TRANSFER QUESTION

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

Position change equals signed area under v-t.

H2
MODEL

Break the graph into simple rectangles/triangles and add their signed areas.

H3
START

The particle starts from x = 0, so its position after 5 s equals the signed area under the given v-t graph from 0 to 5 s. Break the graph into its simple pieces and add their areas. ExamSIDE gives the resulting position as 9 m.

Q5 D2 M4C-C6

EX-M4C-01

QUESTION

A train starts from rest, accelerates uniformly to 80 km/h during time t, then continues at 80 km/h for another 3t. Find the average speed for the whole 4t journey.

TRAIN: ACCELERATE, THEN CRUISE

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE

The first stage has uniform acceleration, so its average speed is the midpoint of 0 and 80.

H2 MODEL

Stage 1 distance = 40t. Stage 2 distance = 80(3t).

H3 START

Stage 1 is uniform acceleration from 0 to 80, so v_{avg},1 = 40 km/h. Hence s_1 = 40t. Stage 2 is constant 80 km/h for 3t, so s_2 = 240t.

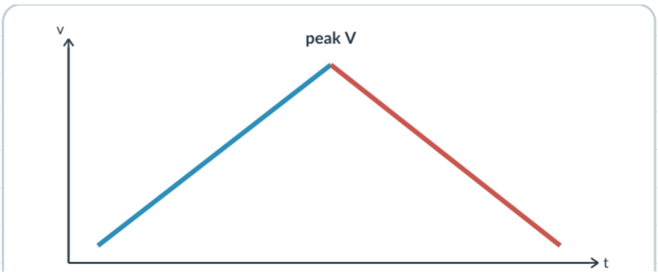
Q6 D4 M4C-C2/C3 EX-M4C-02

QUESTION

A car accelerates from rest at constant rate alpha for some time, then decelerates at constant rate beta to come to rest. If total time is T, express the total distance in terms of alpha, beta and T.

ACCELERATE, THEN BRAKE TO REST

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

Both stages share the same peak velocity V; the v-t graph is one triangle.

H2
MODEL

t_1 = V/alpha and t_2 = V/beta, so T = V(1/alpha + 1/beta).

H3
START

Solve for V, then total distance is the triangle area: S = 1/2 V T.

SET 3 / 8

Stopping, polynomial motion & time intervals

Work first. Hints stay below the work area.

Q1 D2 M4D-C4 **EX-M4D-01**

QUESTION

An automobile travelling at 40 km/h can be stopped in 40 m under a fixed braking condition. Under the same condition, what stopping distance is needed at 80 km/h?

STOPPING DISTANCE WHEN SPEED DOUBLES

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	The braking condition is unchanged, so $\lvert a \rvert$ is the same.
H2 MODEL	Speed doubles: $u_2/u_1 = 2$. Stopping-distance ratio is the square of this speed ratio.
H3 START	For fixed braking acceleration, s_{stop} proportional to u^2. The speed ratio is 80/40 = 2, so the stopping-distance ratio is $2^2 = 4$. Therefore the required distance is 4 x 40 = 160 m.

Q2 D3 M4D-C6 **EX-M4D-02**

QUESTION

A train moving at 20 m/s normally needs 500 m to stop under a fixed retardation. If braking begins only 250 m before the station, the train crosses the station at speed √(x) m/s. Find x.

THE TRAIN BRAKES TOO LATE

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	The first stopping case determines the constant retardation.
H2 MODEL	Use $v^2 = u^2 + 2as$ on the 500 m stop to get a.
H3 START	Reuse the same a over 250 m, then identify x from v^2.

Q3 D3 M4D-H1 EX-M4D-03

QUESTION

A bullet's speed becomes one-third of its entry speed after penetrating 4 cm into a wooden block. Assume constant resistance. If it finally stops after travelling (4 + x) cm in the block, find x.

BULLET PENETRATION UNDER CONSTANT RESISTANCE JEE Main 2022, 27 Jul Evening - adapted

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Constant resistance -> constant retardation. Distance, not time, is the natural variable.
H2 MODEL	Compare the loss in v^2 over the first 4 cm with the remaining loss from $(u/3)^2$ to 0.
H3 START	First loss = $u^2 - u^2/9 = 8u^2/9$ over 4 cm.

Q4 D3 M4D-C3/H1 EX-M4D-04

QUESTION

A train accelerates uniformly. Its engine passes a signal with speed u, and the last carriage passes the same signal with speed v. What is the speed w when the midpoint of the train passes the signal?

SPEED AT THE TRAIN MIDPOINT

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	The midpoint corresponds to half the displacement from engine to last carriage.
H2 MODEL	For full length L: $v^2 = u^2 + 2aL$.
H3 START	For half length L/2: $w^2 = u^2 + aL$. Eliminate aL using the full-length equation.

Q5 D2 M04E

EX-M4E-01

QUESTION

$x(t)=5t^2-4t+5$ m. Find the magnitude of velocity at t=2 s without calculus.

JEE MAIN 2023 - 15 APR MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Compare $x(t)=5t^2-4t+5$ with $x=x_0+ut+(1/2)at^2$.
H2 MODEL	Read $x_0=5$, $u=-4$ and $a=10$ m/s^2.
H3 START	At t=2 s, use $v=u+at = -4 + 10(2)$.

Q6 D2 M04E

EX-M4E-02

QUESTION

$s(t)=2.5t^2$. Find instantaneous speed at t=5 s using the constant-acceleration form.

JEE MAIN 2023 - 13 APR EVENING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Compare $s(t)=2.5t^2$ with $x=x_0+ut+(1/2)at^2$.
H2 MODEL	$u=0$ and $(1/2)a=2.5$, so $a=5$ m/s^2.
H3 START	At t=5 s, $v=u+at = 0 + 5(5)$.

Q7 D2 M04E

EX-M4E-03

QUESTION

Distance from the start is $x=4t^2$. Find velocity at t=5 s under the constant-acceleration interpretation.

JEE MAIN 2023 - 25 JAN EVENING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — Compare $x=4t^2$ with $x=x_0+ut+(1/2)at^2$.

H2 MODEL — u=0 and (1/2)a=4, so a=8 m/s^2.

H3 START — At t=5 s, v=u+at = 8(5).

Q8 D3 M04E

EX-M4E-04

QUESTION

Two buses start together with positions X_P=alpha t+beta t^2 and X_Q=f t-t^2. Find when their velocities are equal.

JEE MAIN 2022 - 25 JUN EVENING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — Read each position polynomial using $x=x_0+ut+(1/2)at^2$.

H2 MODEL — For P: u=alpha, a=2beta. For Q: u=f, a=-2.

H3 START — Set alpha + 2beta t = f - 2t and solve for t.

SET 4 / 8

Intervals, braking pairs & free fall

Work first. Hints stay below the work area.

Q1 D3 M5A-C5

EX-M5A-01

QUESTION

A body travels 102.5 m during its nth second and 115.0 m during its (n+2)th second. Find its constant acceleration.

TWO nth-SECOND DISTANCES

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Both data are interval displacements, not cumulative totals.
H2 MODEL	Use $s_n = u + (a/2)(2n-1)$ for both intervals.
H3 START	Subtract: $115.0 - 102.5 = s_{n+2} - s_n = 2a$.

Q2 D2 M5A-C6

EX-M5A-02

QUESTION

A small toy starts from rest under constant acceleration. It travels 10 m in the first t seconds. How far does it travel in the next t seconds?

THE NEXT t SECONDS

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	"Next t seconds" means the second interval only: S(2t)-S(t).
H2 MODEL	From rest with constant acceleration, S(t) is proportional to t^2. If S(t)=10, then S(2t)=4S(t)=40.
H3 START	Distance in the next t seconds = 40 - 10.

Q3 D3 M-ST05B

EX-M5B-01

QUESTION

A body starts from rest with constant acceleration. It covers S_1 in the first (p-1) s and S_2 in the first p s. In what time is displacement S_1+S_2 covered?

S1 + S2 FROM REST

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

From rest with constant acceleration, write $S(t)=K t^2$ where $K=a/2$.

H2
MODEL

From rest with constant acceleration, write $S(t)=K t^2$ where $K=a/2$. Then $S_1=K(p-1)^2$ and $S_2=Kp^2$.

H3
START

From rest with constant acceleration, write $S(t)=K t^2$ where $K=a/2$. Then $S_1=K(p-1)^2$ and $S_2=Kp^2$. If the same body covers S_1+S_2 in time T, then $K T^2 = K[(p-1)^2+p^2]$.

Q4 D2 M6B-C1/C2

EX-M6B-01

QUESTION

Two cars travel towards each other at 20 m/s each. When they are 300 m apart, both drivers apply brakes and each car retards at 2 m/s^2. Find the distance between them when both come to rest.

TWO CARS BRAKE TOWARDS EACH OTHER

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

Solve each stopping motion independently.

H2
MODEL

Each stopping distance is $u^2/(2|a|)=100$ m.

H3
START

Remaining gap = 300 - 100 - 100.

Q5 D3 M6C-C4 EX-M6C-01

QUESTION

During the one-second interval from t to t+1, a particle has displacement 125 m and its velocity increases by 50 m/s. Find the distance travelled during the (t+2)th second.

DISPLACEMENT + VELOCITY INCREASE

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	The given interval is exactly 1 s, so $\Delta v=50$ immediately gives $a=50\ m/s^2$.
H2 MODEL	Average velocity in the given interval is 125 m/s.
H3 START	Use $(u+v)/2=125$ and $v-u=50$ to recover the endpoint velocities, then advance to the requested later second.

Q6 D2 M7B EX-M7B-01

QUESTION

A ball is dropped from rest from 18 m. Find the height above ground where |v| is numerically equal to |g|, using $g=10\ m/s^2$.

2026-04-05 DROPPED FROM 18 M

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Use $v^2=2gs$ for the distance fallen, then subtract from 18 m.
H2 MODEL	$v=10$, $u=0$. From $v^2=2gs$: $100=20s$, so $s=5$ m fallen.
H3 START	$v=10$, $u=0$. From $v^2=2gs$: $100=20s$, so $s=5$ m fallen. Height above ground = 18-5 = 13 m.

Q7 D3 M7B

EX-M7B-02

QUESTION

A ball is released from height h. Compare the times t_1 and t_2 needed to cover the first and second halves of the distance.

2022-07-29 FIRST HALF VS SECOND HALF

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	For free fall from rest, cumulative time is proportional to √(distance fallen).
H2 MODEL	Let total fall time be T. Time for the first half is t_1=T/√(2).
H3 START	The second-half time is t_2=T-t_1; simplify the ratio t_1:t_2.

Q8 D3 M7B

EX-M7B-03

QUESTION

Drops fall regularly from 9.8 m. When the first drop hits the floor, the third just starts. Locate the second drop from the floor.

2021-08-27 SHOWER DROPS

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Find first-drop fall time, split it into two equal release intervals, then use $s=1/2gt^2$ for the second drop.
H2 MODEL	From h=9.8 m and g=9.8, first-drop time $T=sqrt_2$ s. If the third just begins when the first hits, the release gap is T/2.
H3 START	From h=9.8 m and g=9.8, first-drop time $T=sqrt_2$ s. If the third just begins when the first hits, the release gap is T/2. The second drop has fallen for T/2, so distance fallen = 1/2 $g(T/2)^2$ = 2.45 m.

SET 5 / 8

Free fall to vertical relative motion

Work first. Hints stay below the work area.

Q1 D3 M7C

EX-M7C-01

QUESTION

A body falling under gravity covers points A and B separated by 80 m in 2 s. Find the distance of A from the starting point, using g=10.

2024-01-27 BODY CROSSES A AND B

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Let A be reached at T and B at T+2. Use S(T+2)-S(T)=80, then compute S(T).
H2 MODEL	$S(t)=5t^2$. Use $5[(T+2)^2-T^2]=80$ -> $5(4T+4)=80$ -> T=3 s.
H3 START	$S(t)=5t^2$. Use $5[(T+2)^2-T^2]=80$ -> $5(4T+4)=80$ -> T=3 s. Therefore distance of A from release = S(3)=5*9=45 m.

Q2 D4 M7C

EX-M7C-02

QUESTION

Water droplets fall regularly. The spacing between a droplet observed at its 4th second of fall and the next droplet is 34.3 m. Determine the drop rate, using g=9.8.

2021-07-25 SPACING BETWEEN DROPLETS

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

Write spacing as difference of two free-fall distances with elapsed times separated by the release interval.

H2
MODEL

Let release interval be tau. The older drop has fallen 4 s; the next one has fallen 4-tau s.

H3
START

Let release interval be tau. The older drop has fallen 4 s; the next one has fallen 4-tau s. Spacing = 4.9[16-(4-tau)^2]=34.3.

Q3 D1 M8A-C1

EX-M8A-01

QUESTION

A ball is thrown vertically upward at 150 m/s. With $g=10\ m/s^2$, the ratio of its velocities after 3 s and 5 s is written as $(x+1)/x$. Find x.

VERTICAL VELOCITY AT 3 s AND 5 s

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Upward positive -> a=-g. Both times are before the top.
H2 MODEL	v(3)=120 and v(5)=100, so the ratio is 6/5.
H3 START	v_3=150-10*3=120 m/s and v_5=150-10*5=100 m/s. Ratio = 120/100=6/5=(x+1)/x, giving x=5.

Q4 D2 M8B

EX-M8B-01

QUESTION

A juggler throws n balls per second and releases each new ball when the previous one reaches its highest point. Find the maximum height.

2022-07-29 JUGGLER

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Set t_{up}=1/n, then recover u and H.
H2 MODEL	Time to top = 1/n. Thus u/g=1/n -> u=g/n.
H3 START	Time to top = 1/n. Thus u/g=1/n -> u=g/n. Then $H=u^2/(2g)=g/(2n^2)$.

Q5 D3 M8B

EX-M8B-02

QUESTION

A ball is thrown upward from a tower at 19.6 m/s and hits the ground after 6 s. Find the greatest height above ground reached by the ball.

2022-07-28 BALL FROM TOWER

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	First use the 6 s landing condition to get tower height; then add the rise $u^2/(2g)$.
H2 MODEL	Take upward positive and launch point y=0. At ground after 6 s, y=-h.
H3 START	Take upward positive and launch point y=0. At ground after 6 s, y=-h. So -h=19.6(6)-4.9(36)=117.6-176.4=-58.8, hence tower height h=58.8 m.

Q6 D3 M8C

EX-M8C-01

QUESTION

A ball is thrown vertically upward and reaches maximum height h. Find the ratio of the times at which it is at height h/3 while going up and coming down.

HEIGHT h/3 ON THE WAY UP AND DOWN

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Use $H=u^2/(2g)$, then solve y=h/3 in $y=ut-1/2gt^2$. The two positive roots are the two passage times.
H2 MODEL	At maximum height, $h=u^2/(2g)$. At $y=h/3=u^2/(6g)$, solve $u^2/(6g)=ut-1/2gt^2$.
H3 START	At maximum height, $h=u^2/(2g)$. At $y=h/3=u^2/(6g)$, solve $u^2/(6g)=ut-1/2gt^2$. Multiply by 6g: $u^2=6gut-3g^2t^2$.

Q7 D3 M9A

EX-M9A-01

QUESTION

A ball is projected upward at 50 m/s at t=0. Another is projected upward at 50 m/s at t=2 s. Find the meeting time, g=10.

JEE MAIN 2022, 26 JUN EVENING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	At t=2, Ball 1 is 80 m above launch and moving at 30 m/s. Ball 2 starts at 50 m/s. After that, a_{rel}=0 and closing speed is 20 m/s.
H2 MODEL	At t=2 s, Ball 1: y=50(2)-5(4)=80 m and v=50-20=30 m/s upward. Ball 2 begins at 50 m/s, so relative closing speed is 20 m/s and a_{rel}=0.
H3 START	At t=2 s, Ball 1: y=50(2)-5(4)=80 m and v=50-20=30 m/s upward. Ball 2 begins at 50 m/s, so relative closing speed is 20 m/s and a_{rel}=0. Time after second launch =80/20=4 s.

Q8 D4 M9A

EX-M9A-02

QUESTION

Two equal spherical balls are thrown upward 3 s apart with the same initial velocity 35 m/s. Each radius is 5 cm. Find the collision height, g=10.

JEE MAIN 2021, 26 AUG MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	At the second launch, compute the first ball's state.
H2 MODEL	At the second launch (3 s), Ball 1 is at 60 m and moving upward at 5 m/s. Ball 2 starts upward at 35 m/s, so centre-to-centre closing speed is 30 m/s and remains constant.
H3 START	Ball 2 starts upward at 35 m/s, so centre-to-centre closing speed is 30 m/s and remains constant. Allowing for the 10 cm centre separation at contact gives a collision just before 2 s after the second launch; the corresponding height rounds to the published integer 50 m.

Q1 D2 M9B-C1 EX-M9B-01

QUESTION

A gas balloon rises at 10 m/s. At height 75 m a stone is dropped, while the balloon continues upward at the same speed. Find the balloon's height when the stone hits the ground, taking g=10.

STONE RELEASED FROM RISING BALLOON

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Stone starts with u=+10 m/s in the ground frame. Solve $0=75+10t-5t^2$; then advance the balloon by 10t.
H2 MODEL	Use upward positive. Stone: $y=75+10t-5t^2$.
H3 START	Use upward positive. Stone: $y=75+10t-5t^2$. Setting y=0 gives t=5 s.

Q2 D2 M9C-C1 EX-M9C-01

QUESTION

From a building, Ball A is thrown upward and Ball B downward with the same speed magnitude. Compare their velocities on reaching the ground, neglecting air resistance.

TWO BALLS FROM A BUILDING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	Their velocity directions at impact are both downward. For speed magnitude, use $v^2=u^2+2gH$ for each.
H2 MODEL	Both balls begin with the same speed magnitude u at the same height H above ground. For either ball, the final speed satisfies $v^2=u^2+2gH$.
H3 START	Both balls begin with the same speed magnitude u at the same height H above ground. For either ball, the final speed satisfies $v^2=u^2+2gH$. Therefore the impact speed magnitudes are equal.

Q3 D3 M9D

EX-M9D-01

QUESTION

A body thrown upward from a tower reaches ground in t_1. Thrown downward with the same speed it reaches in t_2. Find the time if simply dropped.

JEE MAIN 2024, 6 APR EVENING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	The upward and downward launches use the same tower height and the same speed magnitude.
H2 MODEL	Use the paired tower equations, or the source-derived relation connecting the two launch times.
H3 START	For the drop-from-rest time, use $t_0 = \sqrt{(t_1\ t_2)}$.

Q4 D2 M9D

EX-M9D-02

QUESTION

From a tower, upward launch reaches ground in 6 s and equal-speed downward launch in 1.5 s. Find the drop-from-rest time.

JEE MAIN 2022, 24 JUN MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	This is the paired-time tower case: upward time 6 s and downward time 1.5 s.
H2 MODEL	Use $t_0 = \sqrt{(t_1\ t_2)} = \sqrt{(6 \times 1.5)}$.
H3 START	$\sqrt{(9)} = 3$ s.

Q5 D3 M10A

EX-M10A-01

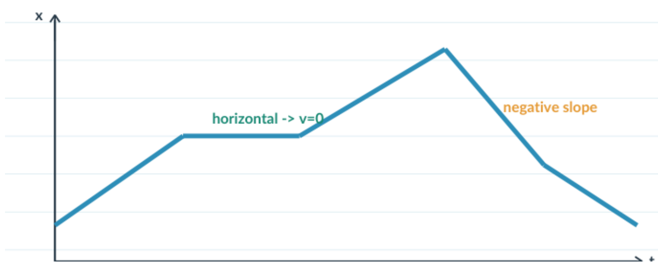
QUESTION

From the supplied displacement-time graph, explain how to obtain average velocity and instantaneous velocity using secant and tangent slopes.

JEE MAIN 2025, 4 APR EVENING

Publication adaptation: original option drawings/statements are not preserved in the audited source; solve the standalone task above.

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	On an x-t graph, height is position and slope is velocity.
H2 MODEL	Average velocity over an interval comes from the secant slope Delta x/Delta t; instantaneous velocity comes from the tangent slope.
H3 START	Test each statement using the appropriate secant or tangent slope, then select the supported statements.

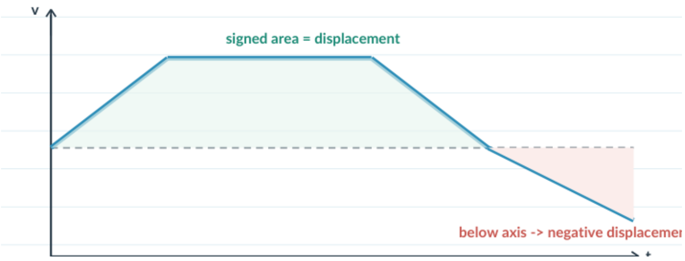
Q6 D3 M10B EX-M10B-01

QUESTION

From a v-t graph over 40 s, find total distance and average velocity.

JEE MAIN 2026, 5 APR EVENING

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — On a v-t graph, signed area is displacement; absolute area is distance.

H2 MODEL — Break the 40 s graph into simple triangular/rectangular regions and keep track of signs.

H3 START — Add absolute areas for distance; add signed areas for displacement; then average velocity = displacement/40 s.

Q7 D2 M10B

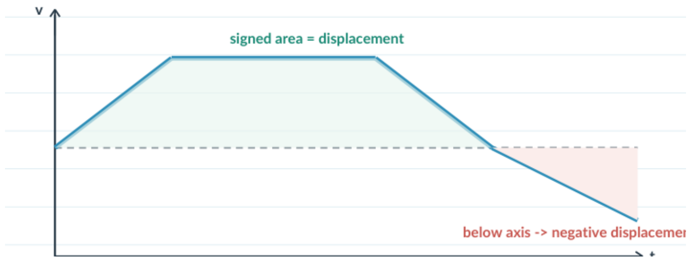
EX-M10B-02

QUESTION

Find distance from t=0 to 4 s from a v-t graph.

JEE MAIN 2025, 28 JAN EVENING

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — Distance from a v-t graph is the area under the curve when velocity stays positive.

H2 MODEL — From 0-2 s use the triangle; from 2-4 s use the rectangle.

H3 START — Area = 10 + 20 m.

Q1 D2 M10B

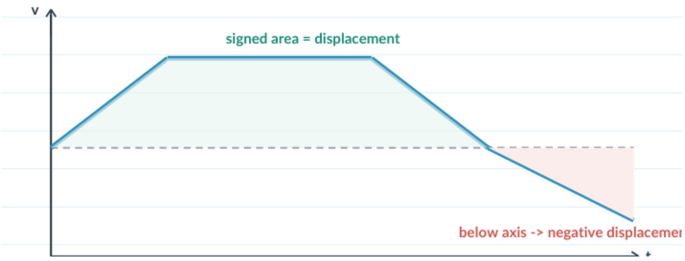
EX-M10B-03

QUESTION

Find airplane distance in the first 30.5 s from its v-t graph.

JEE MAIN 2025, 23 JAN MORNING

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — Treat each simple part of the v-t graph as a geometric area.

H2 MODEL — Add the areas over the first 30.5 s; use the graph units consistently.

H3 START — The source route sums the regions to 12 km.

Q2 D3 M10B

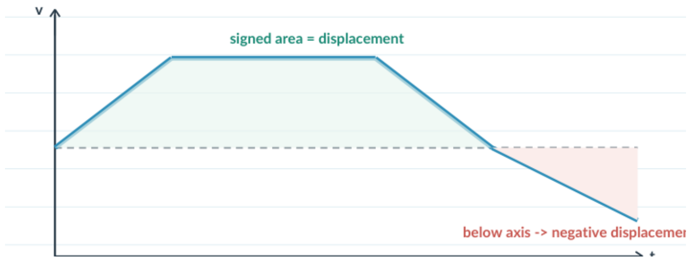
EX-M10B-04

QUESTION

Find displacement : distance from a v-t graph over 0-10 s.

JEE MAIN 2023, 24 JAN EVENING

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE
Signed v-t area gives displacement; absolute area gives distance.

H2
MODEL
Compute the positive and negative regions separately before combining them.

H3
START
The source route gives signed area 16 m and absolute area 48 m; compare 16:48.

Q3 D2 M10B

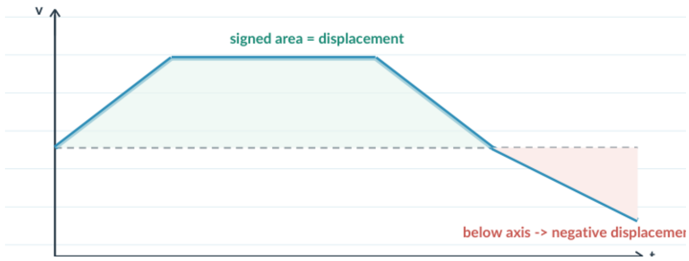
EX-M10B-05

QUESTION

Judge: area under v-t gives distance; area under a-t gives change in velocity.

JEE MAIN 2023, 8 APR EVENING

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

Remember: v-t signed area is displacement, while a-t area is change in velocity.

H2
MODEL

Check Statement I against displacement versus distance; check Statement II against Delta v.

H3
START

The source-supported choice is 'I false, II true'.

Q4 D3 M10B

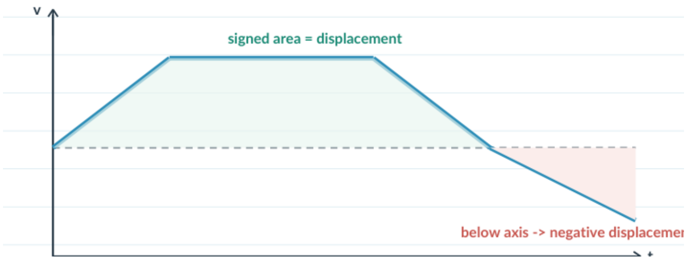
EX-M10B-06

QUESTION

From a v-t graph, find distance/displacement ratio in 25 s.

JEE MAIN 2023, 11 APR MORNING

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE
Use absolute v-t area for distance and signed v-t area for displacement.

H2
MODEL
Evaluate the regions over the 25 s interval before taking the ratio.

H3
START
Distance = 250 m and displacement = 150 m, so form 250/150.

Q5 D2 M10C

EX-M10C-01

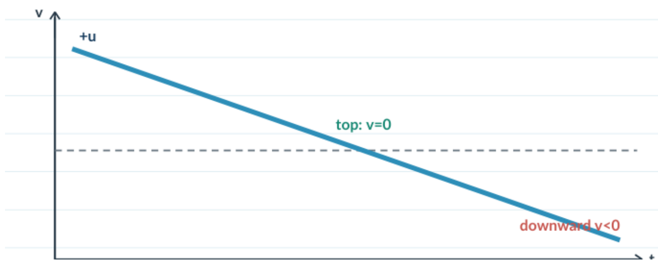
QUESTION

Sketch or describe the v-t graph for a body thrown vertically upward.

JEE MAIN 2017 OFFLINE

Publication adaptation: original option drawings/statements are not preserved in the audited source; solve the standalone task above.

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE
For a vertical throw, acceleration is constant -g, so the v-t graph must be a straight line.

H2
MODEL
Use v = u - gt: the line crosses v=0 at the top and then continues to negative velocity.

H3
START
Choose the graph with constant negative slope through zero and into negative v.

Q6 D3 M10C

EX-M10C-02

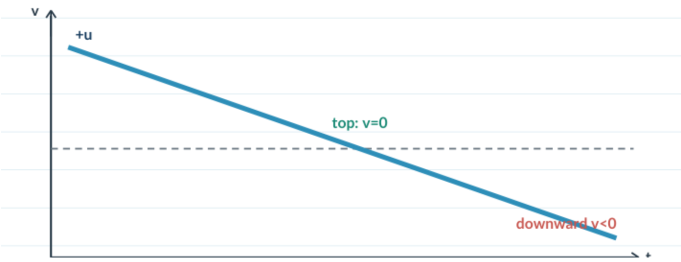
QUESTION

Sketch or describe the v-t curve for a bullet moving downward for 10 s, hitting the ground inelastically, then remaining at rest.

JEE MAIN 2022, 27 JUL MORNING

Publication adaptation: original option drawings/statements are not preserved in the audited source; solve the standalone task above.

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE
Use one sign convention consistently; the source method takes downward as positive.

H2
MODEL
Before impact, v increases linearly from 100 m/s with slope +g.

H3
START
At 10 s the speed is 200 m/s; the inelastic impact makes v jump to 0 and then remain 0.

Q7 D2 M10D

EX-M10D-01

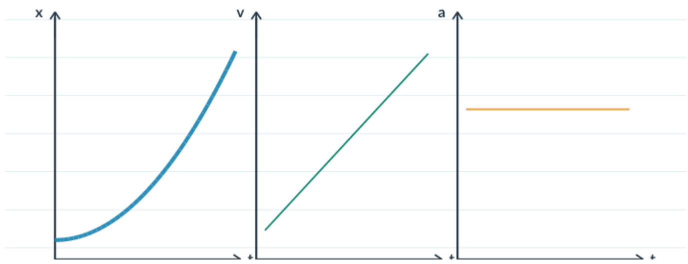
QUESTION

Sketch or describe the x-t, v-t and a-t graph shapes for motion with constant acceleration.

JEE MAIN 2021, 18 MAR MORNING

Publication adaptation: original option drawings/statements are not preserved in the audited source; solve the standalone task above.

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE
Constant acceleration means a-t is horizontal.

H2
MODEL
Then v-t is linear and x-t is quadratic/parabolic.

H3
START
Choose the set showing x-t parabolic, v-t straight and a-t horizontal.

Q8 D3 M10D

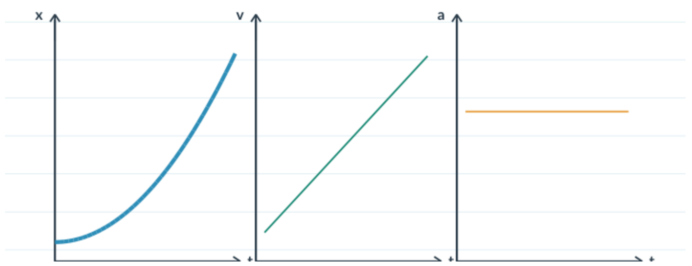
EX-M10D-02

QUESTION

From rest with uniform positive acceleration, identify all correct qualitative graphs.

JEE MAIN 2019, 8 APR EVENING

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — The particle starts from rest with constant positive acceleration.

H2 MODEL — Use v=at and $x=(1/2)at^2$ to infer the graph shapes.

H3 START — Check which candidates show linear v, quadratic x and constant positive a.

Q1 D4 M10E

EX-M10E-01

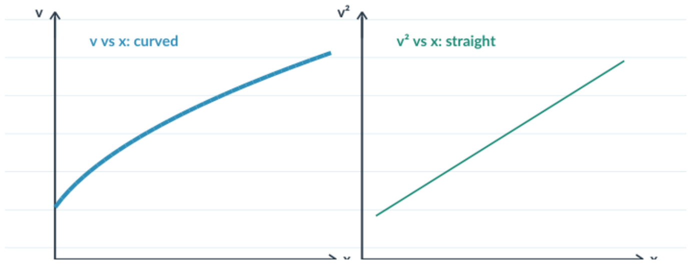
QUESTION

For the given straight-line v-x graph, sketch or describe the corresponding a-x graph.

JEE MAIN 2026, 24 JAN EVENING

Publication adaptation: original option drawings/statements are not preserved in the audited source; solve the standalone task above.

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

A straight v-x relation is not itself constant acceleration.

H2
MODEL

Use the bridge a = v dv/dx. For v=c-mx, dv/dx=-m.

H3
START

Then a=-m(c-mx): a straight line with positive slope and negative intercept.

Q2 D4 M10E

EX-M10E-02

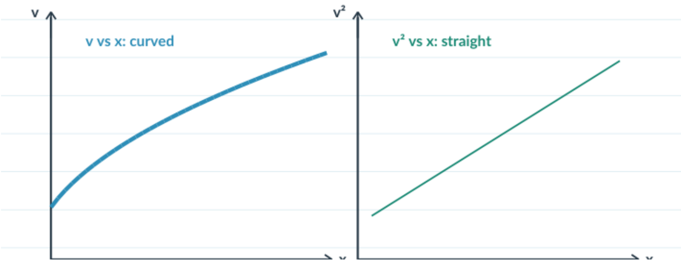
QUESTION

For a linear v-x graph, sketch or describe the corresponding a-x graph.

JEE MAIN 2021, 18 MAR EVENING

Publication adaptation: original option drawings/statements are not preserved in the audited source; solve the standalone task above.

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE — Convert v(x) to a(x) using a = v dv/dx.

H2 MODEL — For a linear v-x relation, dv/dx is constant, so a is proportional to v and is linear in x.

H3 START — The source route gives a straight a-x line with positive slope and negative intercept.

Q3 D2 M10E

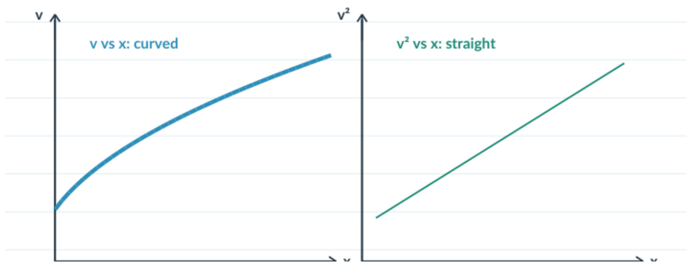
EX-M10E-03

QUESTION

A v^2-x graph is a straight line. Find constant acceleration from its slope.

JEE MAIN 2021, 31 AUG EVENING

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1 NOTICE	For constant acceleration, $v^2 = u^2 + 2ax$ is linear in x.
H2 MODEL	Therefore the slope of a v^2-x graph equals 2a.
H3 START	Given slope 2, solve 2a=2.

Q4 D3 M10E

EX-M10E-04

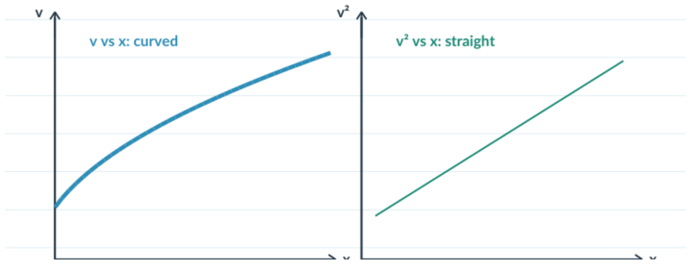
QUESTION

For positive velocity with constant negative acceleration, sketch or describe the v-x graph.

JEE MAIN 2017, 8 APR MORNING

Publication adaptation: original option drawings/statements are not preserved in the audited source; solve the standalone task above.

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE
Use $v^2 = u^2 + 2ax$ with a<0.

H2
MODEL
As x increases, v^2 decreases linearly, but v itself follows a square-root curve.

H3
START
Choose the positive-v graph that decreases nonlinearly with x.

Q5 D4 M10F

EX-M10F-01

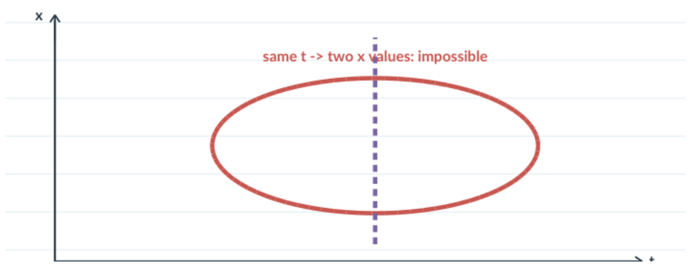
QUESTION

State the single-valued-time test used to decide whether a plotted curve can represent one-dimensional motion. The original option plots are not preserved in the audited source PDF.

JEE MAIN 2025, 3 APR MORNING

Publication adaptation: original option drawings/statements are not preserved in the audited source; solve the standalone task above.

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

For one particle, one instant of time cannot correspond to two positions or two velocities.

H2
MODEL

Check each plotted curve for whether time gives a single-valued state.

H3
START

Reject the curve that assigns two values at the same t; compare the remaining candidates.

Q6 D4 M10F

EX-M10F-02

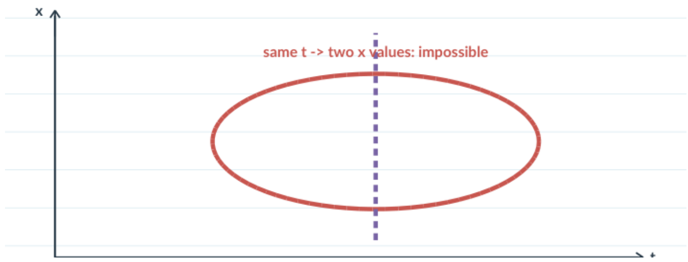
QUESTION

State the consistency test for multiple graph representations of the same motion. The audited source records graph 3 as the inconsistent source option.

JEE MAIN 2018 OFFLINE

Publication adaptation: original option drawings/statements are not preserved in the audited source; solve the standalone task above.

SOURCE-GROUNDED VISUAL



WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE
Translate every candidate graph into the same motion story before comparing them.

H2
MODEL
Compare initial position and especially initial slope, because x-t slope is velocity.

H3
START
The inconsistent graph is the one whose initial slope implies zero initial velocity while the others do not.

Q7 D3 M11D

EX-M11D-01

QUESTION

For motion from rest, the ratio S_{n-1}/S_n of two consecutive one-second distances is written as 1-2/x for n=10. Find x.

JEE MAIN 2024, 5 APR MORNING

WORK HERE - TRY BEFORE LOOKING DOWN

STOP - LOOK BELOW ONLY IF STUCK. READ ONE HINT AT A TIME.

H1
NOTICE

For motion from rest with constant acceleration, nth-second distances are proportional to 2n-1.

H2
MODEL

So s_9:s_{10} = 17:19.

H3
START

Set 17/19 = 1 - 2/x and solve for x.

CHECK YOUR METHOD - FULL DRAFT

MOTION IN A STRAIGHT LINE

Question recap -> method -> answer -> return

USE THIS AFTER ATTEMPTING

Do not start here.

Each card repeats enough of the question so you never have to guess what solution you are reading.

For digital use, each method links back to its question in the combined edition.

CHECK YOUR METHOD - SET 1 / 8

Average speed & average acceleration

Why -> method -> answer -> concept to keep.

EX-M2B-01

Set 1 - Q1

QUESTION RECAP

Half the journey is at 6 m/s. The remaining half is covered in two equal time intervals at 9 m/s and 15 m/s. Find the average speed.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Treat the second half first: equal times at 9 m/s and 15 m/s give 12 m/s for that half. Let each half-distance be L. Then the first-half time is L/6 and the second-half time is L/12. Therefore $v_{avg} = 2L/(L/6 + L/12) = 8$ m/s.

ANSWER / CHECK

8 m/s

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EX-M2B-02

Set 1 - Q2

QUESTION RECAP

A person travels distance x at velocity v_1 and then another equal distance x at velocity v_2 in the same direction. Relate the average velocity v to v_1 and v_2.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Let each leg have distance x. The travel times are x/v_1 and x/v_2, so total distance = 2x and total time = $x/v_1 + x/v_2$. Therefore $v_{avg} = 2x/(x/v_1 + x/v_2) = 2v_1v_2/(v_1+v_2)$.

ANSWER / CHECK

$v = 2v_1v_2/(v_1+v_2)$

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EX-M2B-03

Set 1 - Q3

QUESTION RECAP

An object covers three equal consecutive distances at speeds v_1, v_2 and v_3. Find the average speed.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Let each equal segment be L. The three times are L/v_1, L/v_2 and L/v_3, so total time = $L(1/v_1 + 1/v_2 + 1/v_3)$ and total distance = 3L. Dividing gives $v_{avg} = 3/(1/v_1 + 1/v_2 + 1/v_3)$.

ANSWER / CHECK

$v_{avg} = 3/(1/v_1 + 1/v_2 + 1/v_3)$

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EX-M2B-04

Set 1 - Q4

QUESTION RECAP

A vehicle travels 4 km at 3 km/h and another 4 km at 5 km/h. Find the average speed.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Do not average 3 and 5 directly because the legs have equal distance, not equal time. The times are 4/3 h and 4/5 h. Thus total time = 32/15 h and v_{avg} = 8/(32/15) = 3.75 km/h.

ANSWER / CHECK

3.75 km/h

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

CHECK YOUR METHOD - SET 1 / 8

Average speed & average acceleration

Why -> method -> answer -> concept to keep.

EX-M2B-05

Set 1 - Q5

QUESTION RECAP

A car travels distance x at speed v_1 and then the same distance x at speed v_2. Find average speed.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Let each leg be x. Total distance = 2x, while total time = $x/v_1 + x/v_2$. Hence $v_{avg} = 2x/(x/v_1 + x/v_2) = 2v_1v_2/(v_1+v_2)$.

ANSWER / CHECK

$\mathbf{v_{avg} = 2v_1v_2/(v_1+v_2)}$

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EX-M2B-06

Set 1 - Q6

QUESTION RECAP

A person moves distance x at 5 m/s and the next 3x/2 at speed v_2. The overall average velocity is 50/7 m/s. Find v_2.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Write time for each leg as distance/speed. Total displacement = x + 3x/2 = 5x/2, while total time = $x/5 + 3x/(2v_2)$. Set average velocity = $(5x/2)/(x/5 + 3x/(2v_2)) = 50/7$; cancel x and solve to get $v_2 = 10$ m/s.

ANSWER / CHECK

$\mathbf{v_2=10\ m/s}$

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EX-M2B-07

Set 1 - Q7

QUESTION RECAP

A horse rider covers half the distance at 5 m/s. The remaining half is covered at 10 m/s and 15 m/s for equal times. The mean speed is x/7 m/s. Find x.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Let total distance be D. First-half time=D/10. For the second half, (10+15)tau=D/2, so tau=D/50 and second-half time=D/25. Average=50/7 m/s, hence x=50.

ANSWER / CHECK

x=50

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

EX-M2B-08

Set 1 - Q8

QUESTION RECAP

A car covers three equal one-third portions at 11, 22 and 33 m/s. Find average velocity.

WHY THIS WORKS

Average speed or velocity must be built from total distance/displacement divided by total time. Different speeds contribute through the time spent at each speed, not by a blind arithmetic mean.

METHOD

Use one common distance L for each third. Total distance = 3L and total time = L/11 + L/22 + L/33. Therefore v_{avg} = 3/(1/11 + 1/22 + 1/33) = 18 m/s.

ANSWER / CHECK

18 m/s

CONCEPT TO KEEP

Keep: average speed = total distance / total time. Equal-distance legs are time-weighted; for two equal distances this becomes the harmonic-mean form $2v_1v_2/(v_1+v_2)$.

CHECK YOUR METHOD - SET 1 / 8

Average speed & average acceleration

EX-M3A-01

Set 1 - Q9

QUESTION RECAP

A tennis ball is dropped from 9.8 m and rebounds to 5.0 m. Contact with the floor lasts 0.2 s. Take $g=10\ m/s^2$. Find average acceleration during contact.

WHY THIS WORKS

Acceleration depends on change in velocity, so direction matters. A rebound reverses the velocity sign and makes the velocity change larger than the change in speed alone.

METHOD

Choose upward positive. Just before impact v=-14 m/s; just after rebound v=+10 m/s. Delta v=24 m/s in 0.2 s -> average acceleration=120 m/s^2 upward.

ANSWER / CHECK

Delta v=24 m/s in 0.2 s -> average acceleration=120 m/s^2 upward

CONCEPT TO KEEP

Keep: average acceleration = Δv/Δt, with velocity signs included.

CHECK YOUR METHOD - SET 2 / 8

Uniform acceleration: equations & stages

Why -> method -> answer -> concept to keep.

EX-M4A-01

Set 2 - Q1

QUESTION RECAP

A particle moves in +x with u = 10 m/s and constant a = 2 m/s^2. How long to reach 60 m/s?

WHY THIS WORKS

Here distance is not needed: the known quantities are initial velocity, final velocity and constant acceleration, so the direct velocity-time relation is the efficient model.

METHOD

u = 10, v = 60, a = +2. Use v = u + at: 60 = 10 + 2t -> t = 25 s. Check: positive acceleration increases velocity.

ANSWER / CHECK

25 s

CONCEPT TO KEEP

Keep: if u, v, a and t are the useful variables under constant acceleration, v = u + at is the direct relation.

EX-M4B-01

Set 2 - Q2

QUESTION RECAP

A bus moving at 72 km/h is brought to rest uniformly in 4 s. Find the braking distance.

WHY THIS WORKS

Uniform braking is one constant-acceleration interval. First recover the acceleration from the velocity change, then use it consistently to find the distance.

METHOD

Convert 72 km/h = 20 m/s. With v = 0 and t = 4 s, 0 = 20 + 4a -> a = -5 m/s^2. Then s = ut + $1/2at^2$ = 20(4) + 1/2(-5)(16) = 80 - 40 = 40 m.

ANSWER / CHECK

40 m

CONCEPT TO KEEP

Keep: one constant-acceleration model must be used consistently through a uniform braking interval.

EX-M4B-03

Set 2 - Q3

QUESTION RECAP

A body starts from rest with constant acceleration. It covers S_1 in the first (p-1) s and S_2 in the first p s. In what time is displacement S_1+S_2 covered?

WHY THIS WORKS

From rest under constant acceleration, cumulative displacement is proportional to t^2. That lets two cumulative distances be combined without first solving for the acceleration.

METHOD

Let cumulative displacement from rest be Kt^2, where K = 1/2a. Then $S_1 = K(p-1)^2$ and $S_2 = Kp^2$. If $S_1 + S_2$ is covered in time T, $KT^2 = K[(p-1)^2 + p^2]$. Thus $T^2 = 2p^2 - 2p + 1$, so $T = \sqrt{(2p^2 - 2p + 1)}$ s.

ANSWER / CHECK

$T = \sqrt{(2p^2 - 2p + 1)}$ s

CONCEPT TO KEEP

Keep: from rest at constant acceleration, $S(t) \propto t^2$.

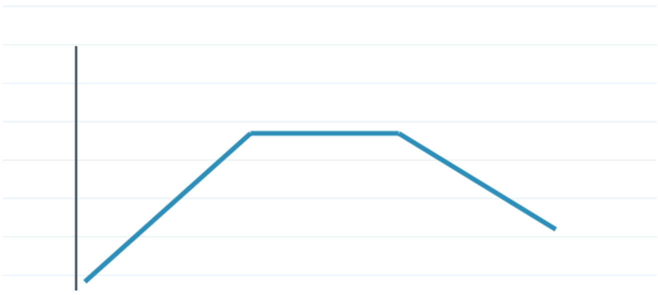
EX-M4B-04

Set 2 - Q4

QUESTION RECAP

A particle starts at the origin and moves along +x with the shown v-t graph. Find its position at t = 5 s.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

On a velocity-time graph, signed area gives displacement. Because the particle starts at the origin, its final position equals that accumulated signed area.

METHOD

The particle starts from x = 0, so its position after 5 s equals the signed area under the given v-t graph from 0 to 5 s. Break the graph into its simple pieces and add their areas. ExamSIDE gives the resulting position as 9 m.

ANSWER / CHECK

the resulting position as 9 m

CONCEPT TO KEEP

Keep: signed area under v-t = displacement; add the initial position to obtain final position.

EX-M4C-01

Set 2 - Q5

QUESTION RECAP

A train starts from rest, accelerates uniformly to 80 km/h during time t, then continues at 80 km/h for another 3t. Find the average speed for the whole 4t journey.

WHY THIS WORKS

A multistage journey must be handled stage by stage. The accelerating stage has a linearly changing velocity, while the cruising stage has constant velocity.

METHOD

Stage 1 is uniform acceleration from 0 to 80, so v_{avg},1 = 40 km/h. Hence s_1 = 40t. Stage 2 is constant 80 km/h for 3t, so s_2 = 240t. Total distance = 280t and total time = 4t. Therefore average speed = 70 km/h.

ANSWER / CHECK

average speed = 70 km/h

CONCEPT TO KEEP

Keep: for staged motion, compute each stage distance and then use total distance / total time.

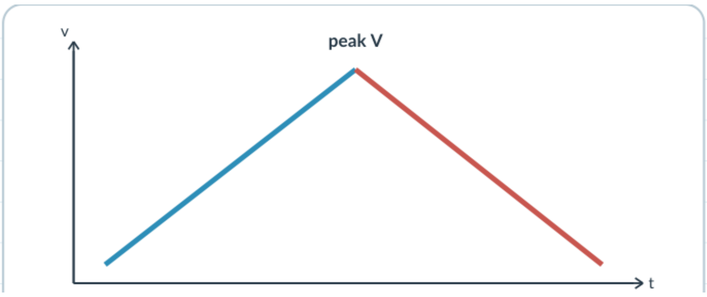
EX-M4C-02

Set 2 - Q6

QUESTION RECAP

A car accelerates from rest at constant rate alpha for some time, then decelerates at constant rate beta to come to rest. If total time is T, express the total distance in terms of alpha, beta and T.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

The two stages meet at one common peak velocity. In a v-t picture the whole journey is one triangle, so time relations determine the peak and triangle area gives distance.

METHOD

Let V be the peak velocity. During acceleration, t_1 = V/alpha. During braking, t_2 = V/beta. Thus T = V(1/alpha + 1/beta), so V = alpha beta T / (alpha + beta). The complete v-t graph is a triangle, therefore S = 1/2 V T. Hence S = alpha beta T^2 / [2(alpha + beta)].

ANSWER / CHECK

S = alpha beta T^2 / [2(alpha + beta)]

CONCEPT TO KEEP

Keep: a linear accelerate-then-brake v-t graph forms a triangle; its area is total displacement.

CHECK YOUR METHOD - SET 3 / 8

Stopping, polynomial motion & time intervals

Why -> method -> answer -> concept to keep.

EX-M4D-01

Set 3 - Q1

QUESTION RECAP

An automobile travelling at 40 km/h can be stopped in 40 m under a fixed braking condition. Under the same condition, what stopping distance is needed at 80 km/h?

WHY THIS WORKS

When acceleration is constant and time is not the useful variable, $v^2 = u^2 + 2as$ connects speed directly with distance. This also exposes stopping-distance scaling with u^2.

METHOD

For fixed braking acceleration, s_{stop} proportional to u^2. The speed ratio is 80/40 = 2, so the stopping-distance ratio is $2^2 = 4$. Therefore the required distance is 4 x 40 = 160 m.

ANSWER / CHECK

160 m

CONCEPT TO KEEP

Keep: with fixed braking acceleration, stopping distance is proportional to the square of entry speed.

EX-M4D-02

Set 3 - Q2

QUESTION RECAP

A train moving at 20 m/s normally needs 500 m to stop under a fixed retardation. If braking begins only 250 m before the station, the train crosses the station at speed √(x) m/s. Find x.

WHY THIS WORKS

When acceleration is constant and time is not the useful variable, $v^2 = u^2 + 2as$ connects speed directly with distance. This also exposes stopping-distance scaling with u^2.

METHOD

Full stop: $0 = 20^2 + 2a(500)$, so 1000a = -400 and a = -0.4 m/s^2. If braking starts 250 m before the station: $v^2 = 20^2 + 2(-0.4)(250) = 400 - 200 = 200$. Since v = sqrtx, x = 200.

ANSWER / CHECK

Since v = sqrtx, x = 200

CONCEPT TO KEEP

Keep: with fixed braking acceleration, stopping distance is proportional to the square of entry speed.

CHECK YOUR METHOD - SET 3 / 8

Stopping, polynomial motion & time intervals

Why -> method -> answer -> concept to keep.

EX-M4D-03

Set 3 - Q3

QUESTION RECAP

A bullet's speed becomes one-third of its entry speed after penetrating 4 cm into a wooden block. Assume constant resistance. If it finally stops after travelling (4 + x) cm in the block, find x.

WHY THIS WORKS

When acceleration is constant and time is not the useful variable, $v^2 = u^2 + 2as$ connects speed directly with distance. This also exposes stopping-distance scaling with u^2.

METHOD

Let the entry speed be u and constant retardation magnitude be b. Over the first 4 cm, the loss in v^2 is $u^2 - (u/3)^2 = 8u^2/9$. Over the remaining distance x, the loss is $(u/3)^2 - 0 = u^2/9$. Because the loss in v^2 per unit distance is constant, distance is proportional to the v^2 loss. Therefore x/4 = (1/9)/(8/9) = 1/8, giving x = 0.5 cm.

ANSWER / CHECK

x/4 = (1/9)/(8/9) = 1/8, giving x = 0.5 cm

CONCEPT TO KEEP

Keep: with fixed braking acceleration, stopping distance is proportional to the square of entry speed.

EX-M4D-04

Set 3 - Q4

QUESTION RECAP

A train accelerates uniformly. Its engine passes a signal with speed u, and the last carriage passes the same signal with speed v. What is the speed w when the midpoint of the train passes the signal?

WHY THIS WORKS

When acceleration is constant and time is not the useful variable, $v^2 = u^2 + 2as$ connects speed directly with distance. This also exposes stopping-distance scaling with u^2.

METHOD

Let train length be L. When the engine passes the signal, speed is u. When the last carriage passes, the engine has advanced L, so $v^2 = u^2 + 2aL$. At the midpoint passage, displacement is L/2, so $w^2 = u^2 + 2a(L/2) = u^2 + aL$. From the full-length equation, $aL = (v^2 - u^2)/2$. Therefore $w^2 = (u^2 + v^2)/2$ and $w = \text{sqrt}[(u^2+v^2)/2]$.

ANSWER / CHECK

$w^2 = (u^2 + v^2)/2$ and $w = \text{sqrt}[(u^2+v^2)/2]$

CONCEPT TO KEEP

Keep: with fixed braking acceleration, stopping distance is proportional to the square of entry speed.

CHECK YOUR METHOD - SET 3 / 8

Stopping, polynomial motion & time intervals

Why -> method -> answer -> concept to keep.

EX-M4E-01

Set 3 - Q5

QUESTION RECAP

$x(t)=5t^2-4t+5$ m. Find the magnitude of velocity at t=2 s without calculus.

WHY THIS WORKS

A quadratic position-time expression can be matched term-by-term with $x = x_0 + ut + 1/2\ at^2$. This reveals u and a without using calculus.

METHOD

Compare with $x=x_0+ut+1/2at^2$: $x_0=5$, $u=-4$, $a=10$. Then $v=u+at=-4+20=16$ m/s.

ANSWER / CHECK

16 m/s

CONCEPT TO KEEP

Keep: match a quadratic x(t) to $x_0 + ut + 1/2\ at^2$ to read u and a directly.

EX-M4E-02

Set 3 - Q6

QUESTION RECAP

$s(t)=2.5t^2$. Find instantaneous speed at t=5 s using the constant-acceleration form.

WHY THIS WORKS

A quadratic position-time expression can be matched term-by-term with $x = x_0 + ut + 1/2\ at^2$. This reveals u and a without using calculus.

METHOD

Read $u=0$ and $1/2a=2.5 \rightarrow a=5$ m/s^2. Then $v=at=25$ m/s.

ANSWER / CHECK

25 m/s

CONCEPT TO KEEP

Keep: match a quadratic x(t) to $x_0 + ut + 1/2\ at^2$ to read u and a directly.

CHECK YOUR METHOD - SET 3 / 8

Stopping, polynomial motion & time intervals

Why -> method -> answer -> concept to keep.

EX-M4E-03

Set 3 - Q7

QUESTION RECAP

Distance from the start is $x=4t^2$. Find velocity at t=5 s under the constant-acceleration interpretation.

WHY THIS WORKS

A quadratic position-time expression can be matched term-by-term with $x = x_0 + ut + 1/2\ at^2$. This reveals u and a without using calculus.

METHOD

Read u=0 and 1/2a=4 -> a=8 m/s^2. Then v=8t=40 m/s at 5 s.

ANSWER / CHECK

40 m/s

CONCEPT TO KEEP

Keep: match a quadratic x(t) to $x_0 + ut + 1/2\ at^2$ to read u and a directly.

EX-M4E-04

Set 3 - Q8

QUESTION RECAP

Two buses start together with positions X_P=alpha t+beta t^2 and X_Q=f t-t^2. Find when their velocities are equal.

WHY THIS WORKS

A quadratic position-time expression can be matched term-by-term with $x = x_0 + ut + 1/2\ at^2$. This reveals u and a without using calculus.

METHOD

Read u_P=alpha, a_P=2beta; u_Q=f, a_Q=-2. Set alpha+2beta t=f-2t, so t=(f-alpha)/[2(beta+1)].

ANSWER / CHECK

t = (f - alpha)/[2(beta + 1)]

CONCEPT TO KEEP

Keep: match a quadratic x(t) to $x_0 + ut + 1/2\ at^2$ to read u and a directly.

CHECK YOUR METHOD - SET 4 / 8

Intervals, braking pairs & free fall

Why -> method -> answer -> concept to keep.

EX-M5A-01

Set 4 - Q1

QUESTION RECAP

A body travels 102.5 m during its nth second and 115.0 m during its (n+2)th second. Find its constant acceleration.

WHY THIS WORKS

For constant acceleration, nth-second questions are interval questions built from cumulative displacement. Subtracting nearby cumulative expressions removes quantities that are not needed.

METHOD

For constant acceleration, $s_n = u + (a/2)(2n-1)$. Two seconds later, $s_{n+2} = u + (a/2)(2n+3)$. Subtracting cancels u and n: $s_{n+2} - s_n = 2a$. Therefore 115.0 - 102.5 = 12.5 = 2a, so a = 6.25 m/s^2.

ANSWER / CHECK

6.25 m/s^2

CONCEPT TO KEEP

Keep: nth-second displacement comes from subtracting cumulative displacements at adjacent times.

EX-M5A-02

Set 4 - Q2

QUESTION RECAP

A small toy starts from rest under constant acceleration. It travels 10 m in the first t seconds. How far does it travel in the next t seconds?

WHY THIS WORKS

For constant acceleration, nth-second questions are interval questions built from cumulative displacement. Subtracting nearby cumulative expressions removes quantities that are not needed.

METHOD

Starting from rest with constant acceleration, $S(t)=1/2\ at^2$. Given S(t)=10 m. Then $S(2t)=1/2\ a(2t)^2 = 4S(t)=40$ m. The question asks only for the next t-second interval, so subtract: 40 - 10 = 30 m.

ANSWER / CHECK

30 m

CONCEPT TO KEEP

Keep: nth-second displacement comes from subtracting cumulative displacements at adjacent times.

CHECK YOUR METHOD - SET 4 / 8

Intervals, braking pairs & free fall

Why -> method -> answer -> concept to keep.

EX-M5B-01

Set 4 - Q3

QUESTION RECAP

A body starts from rest with constant acceleration. It covers S_1 in the first (p-1) s and S_2 in the first p s. In what time is displacement S_1+S_2 covered?

WHY THIS WORKS

Because the body starts from rest with constant acceleration, cumulative displacement has the common form Kt^2. The common factor K cancels, so the problem is really about time-squared scaling.

METHOD

From rest with constant acceleration, write $S(t)=Kt^2$ where $K=a/2$. Then $S_1=K(p-1)^2$ and $S_2=Kp^2$. If S_1+S_2 is covered in time T, then $KT^2=K[(p-1)^2+p^2]$. Cancel K to obtain $T^2=2p^2-2p+1$, so $T=\sqrt{(2p^2-2p+1)}$ s.

ANSWER / CHECK

$T = \sqrt{(2p^2 - 2p + 1)}$ s

CONCEPT TO KEEP

Keep: when displacement from rest scales as t^2, combine the squared times before taking the square root.

EX-M6B-01

Set 4 - Q4

QUESTION RECAP

Two cars travel towards each other at 20 m/s each. When they are 300 m apart, both drivers apply brakes and each car retards at 2 m/s^2. Find the distance between them when both come to rest.

WHY THIS WORKS

Both cars have the same braking model. Find each stopping distance, then compare the combined distance used with the original separation.

METHOD

Each car has u=20 m/s and deceleration magnitude 2 m/s^2. For one car, $s_{stop}=u^2/(2|a|)=400/4=100$ m. Both together consume 200 m of the original 300 m separation. Therefore the distance between them at rest is 100 m.

ANSWER / CHECK

the distance between them at rest is 100 m

CONCEPT TO KEEP

Keep: stopping-distance comparisons are easiest when each body is solved under the same braking model.

EX-M6C-01

Set 4 - Q5

QUESTION RECAP

During the one-second interval from t to t+1, a particle has displacement 125 m and its velocity increases by 50 m/s. Find the distance travelled during the (t+2)th second.

WHY THIS WORKS

A one-second displacement equals the average velocity during that second times one second. With constant acceleration, the velocity change also fixes the acceleration.

METHOD

The interval length is 1 s. Therefore a=Delta v/Delta t=50 m/s^2. Its displacement is 125 m, so average velocity over that interval is 125 m/s. Let the velocity at time t be u; at t+1 it is u+50. Then [u+(u+50)]/2=125, giving u=100 m/s and v(t+1)=150 m/s. The requested (t+2)th second is the next one-second interval, starting at 150 m/s with a=50 m/s^2. Its displacement is 150(1)+1/2(50)(1)^2=175 m.

ANSWER / CHECK

Its displacement is 150(1)+1/2(50)(1)^2=175 m

CONCEPT TO KEEP

Keep: over a 1 s interval, displacement equals the interval average velocity; Δv over that second gives a.

EX-M7B-01

Set 4 - Q6

QUESTION RECAP

A ball is dropped from rest from 18 m. Find the height above ground where |v| is numerically equal to |g|, using g=10 m/s^2.

WHY THIS WORKS

Free fall from rest is constant-acceleration motion. Use cumulative displacement carefully: equal fractions of distance do not take equal fractions of time.

METHOD

v=10, u=0. From v^2=2gs: 100=20s, so s=5 m fallen. Height above ground = 18-5 = 13 m.

ANSWER / CHECK

Height above ground = 18-5 = 13 m

CONCEPT TO KEEP

Keep: in free fall from rest, distance ∝ t^2, so equal distances do not correspond to equal times.

EX-M7B-02

Set 4 - Q7

QUESTION RECAP

A ball is released from height h. Compare the times t_1 and t_2 needed to cover the first and second halves of the distance.

WHY THIS WORKS

Free fall from rest is constant-acceleration motion. Use cumulative displacement carefully: equal fractions of distance do not take equal fractions of time.

METHOD

Let total fall time be T. First half: t_1=T/$sqrt_2$. Hence t_2=T-t_1=T(1-1/$sqrt_2$). Therefore t_2=($sqrt_2$-1)t_1 or equivalently t_1=($sqrt_2$+1)t_2.

ANSWER / CHECK

t_2=($sqrt_2$-1)t_1 or equivalently t_1=($sqrt_2$+1)t_2

CONCEPT TO KEEP

Keep: in free fall from rest, distance ∝ t^2, so equal distances do not correspond to equal times.

EX-M7B-03

Set 4 - Q8

QUESTION RECAP

Drops fall regularly from 9.8 m. When the first drop hits the floor, the third just starts. Locate the second drop from the floor.

WHY THIS WORKS

Free fall from rest is constant-acceleration motion. Use cumulative displacement carefully: equal fractions of distance do not take equal fractions of time.

METHOD

From h=9.8 m and g=9.8, first-drop time T=$sqrt_2$ s. If the third just begins when the first hits, the release gap is T/2. The second drop has fallen for T/2, so distance fallen = 1/2 g$(T/2)^2$ = 2.45 m. Height above floor = 9.8-2.45 = 7.35 m.

ANSWER / CHECK

Height above floor = 9.8-2.45 = 7.35 m

CONCEPT TO KEEP

Keep: in free fall from rest, distance ∝ t^2, so equal distances do not correspond to equal times.

CHECK YOUR METHOD - SET 5 / 8

Free fall to vertical relative motion

Why -> method -> answer -> concept to keep.

EX-M7C-01

Set 5 - Q1

QUESTION RECAP

A body falling under gravity covers points A and B separated by 80 m in 2 s. Find the distance of A from the starting point, using g=10.

WHY THIS WORKS

For freely falling drops, every object follows the same $y \propto t^2$ law but with a different clock start. Translate the spacing into a difference of cumulative falls.

METHOD

$S(t)=5t^2$. Use $5[(T+2)^2-T^2]=80$ -> $5(4T+4)=80$ -> T=3 s. Therefore distance of A from release = S(3)=5*9=45 m.

ANSWER / CHECK

distance of A from release = S(3)=5*9=45 m

CONCEPT TO KEEP

Keep: regularly released drops share the same fall law but have shifted time origins.

EX-M7C-02

Set 5 - Q2

QUESTION RECAP

Water droplets fall regularly. The spacing between a droplet observed at its 4th second of fall and the next droplet is 34.3 m. Determine the drop rate, using g=9.8.

WHY THIS WORKS

For freely falling drops, every object follows the same $y \propto t^2$ law but with a different clock start. Translate the spacing into a difference of cumulative falls.

METHOD

Let release interval be tau. The older drop has fallen 4 s; the next one has fallen 4-tau s. Spacing = $4.9[16-(4-\text{tau})^2]=34.3$. Divide by 4.9: $16-(4-\text{tau})^2=7$, so $(4-\text{tau})^2=9$ and tau=1 s. Therefore the rate is 1 drop/s.

ANSWER / CHECK

the rate is 1 drop/s

CONCEPT TO KEEP

Keep: regularly released drops share the same fall law but have shifted time origins.

EX-M8A-01

Set 5 - Q3

QUESTION RECAP

A ball is thrown vertically upward at 150 m/s. With $g=10\ m/s^2$, the ratio of its velocities after 3 s and 5 s is written as (x+1)/x. Find x.

WHY THIS WORKS

For vertical motion with upward positive, velocity changes linearly as v = u - gt. Evaluate the velocity at each requested time before forming the ratio.

METHOD

v_3=150-10*3=120 m/s and v_5=150-10*5=100 m/s. Ratio = 120/100=6/5=(x+1)/x, giving x=5.

ANSWER / CHECK

x=5

CONCEPT TO KEEP

Keep: for upward-positive vertical motion, v = u - gt.

EX-M8B-01

Set 5 - Q4

QUESTION RECAP

A juggler throws n balls per second and releases each new ball when the previous one reaches its highest point. Find the maximum height.

WHY THIS WORKS

The highest point is identified by v = 0. The time to the top and the maximum-height relation connect the release timing to the launch speed and height.

METHOD

Time to top = 1/n. Thus u/g=1/n -> u=g/n. Then $H=u^2/(2g)=g/(2n^2)$.

ANSWER / CHECK

Then $H=u^2/(2g)=g/(2n^2)$

CONCEPT TO KEEP

Keep: at the highest point v = 0, so time-to-top and maximum height follow directly from constant-gravity relations.

EX-M8B-02

Set 5 - Q5

QUESTION RECAP

A ball is thrown upward from a tower at 19.6 m/s and hits the ground after 6 s. Find the greatest height above ground reached by the ball.

WHY THIS WORKS

The highest point is identified by v = 0. The time to the top and the maximum-height relation connect the release timing to the launch speed and height.

METHOD

Take upward positive and launch point y=0. At ground after 6 s, y=-h. So -h=19.6(6)-4.9(36)=117.6-176.4=-58.8, hence tower height h=58.8 m. Rise above launch point is $u^2/(2g)$=19.6 m. Greatest height above ground = 78.4 m.

ANSWER / CHECK

Greatest height above ground = 78.4 m

CONCEPT TO KEEP

Keep: at the highest point v = 0, so time-to-top and maximum height follow directly from constant-gravity relations.

EX-M8C-01

Set 5 - Q6

QUESTION RECAP

A ball is thrown vertically upward and reaches maximum height h. Find the ratio of the times at which it is at height h/3 while going up and coming down.

WHY THIS WORKS

The same height is reached twice because the upward and downward parts of the trajectory are symmetric in the constant-gravity model. The two roots of the position equation are the two times.

METHOD

At maximum height, $h=u^2/(2g)$. At $y=h/3=u^2/(6g)$, solve $u^2/(6g)=ut-1/2gt^2$. Multiply by 6g: $u^2=6gut-3g^2t^2$. Let z=gt/u. Then $3z^2-6z+1=0$, so $z=1\pm sqrt_6/3$. Therefore $t_{up}:t_{down}=(1-sqrt_6/3):(1+sqrt_6/3)$, equivalent to $(sqrt_3-sqrt_2)/(sqrt_3+sqrt_2)$.

ANSWER / CHECK

$t_{up}:t_{down}=(1-sqrt_6/3):(1+sqrt_6/3)$, equivalent to $(sqrt_3-sqrt_2)/(sqrt_3+sqrt_2)$

CONCEPT TO KEEP

Keep: a height below the maximum is normally crossed twice - once upward and once downward.

CHECK YOUR METHOD - SET 5 / 8

Free fall to vertical relative motion

Why -> method -> answer -> concept to keep.

EX-M9A-01

Set 5 - Q7

QUESTION RECAP

A ball is projected upward at 50 m/s at t=0. Another is projected upward at 50 m/s at t=2 s. Find the meeting time, g=10.

WHY THIS WORKS

For delayed launches, use one common global clock or reset the clock explicitly at the second launch. After both are in flight, relative motion can simplify the meeting calculation.

METHOD

At t=2 s, Ball 1: y=50(2)-5(4)=80 m and v=50-20=30 m/s upward. Ball 2 begins at 50 m/s, so relative closing speed is 20 m/s and a_{rel}=0. Time after second launch =80/20=4 s. Global meeting time = 6 s.

ANSWER / CHECK

Global meeting time = 6 s

CONCEPT TO KEEP

Keep: delayed-start problems need a common clock before comparing positions or using relative motion.

EX-M9A-02

Set 5 - Q8

QUESTION RECAP

Two equal spherical balls are thrown upward 3 s apart with the same initial velocity 35 m/s. Each radius is 5 cm. Find the collision height, g=10.

WHY THIS WORKS

For delayed launches, use one common global clock or reset the clock explicitly at the second launch. After both are in flight, relative motion can simplify the meeting calculation.

METHOD

At the second launch (3 s), Ball 1 is at 60 m and moving upward at 5 m/s. Ball 2 starts upward at 35 m/s, so centre-to-centre closing speed is 30 m/s and remains constant. Allowing for the 10 cm centre separation at contact gives a collision just before 2 s after the second launch; the corresponding height rounds to the published integer 50 m.

ANSWER / CHECK

a collision just before 2 s after the second launch; the corresponding height rounds to the published integer 50 m

CONCEPT TO KEEP

Keep: delayed-start problems need a common clock before comparing positions or using relative motion.

CHECK YOUR METHOD - SET 6 / 8

Paired launches & first graph reading

Why -> method -> answer -> concept to keep.

EX-M9B-01

Set 6 - Q1

QUESTION RECAP

A gas balloon rises at 10 m/s. At height 75 m a stone is dropped, while the balloon continues upward at the same speed. Find the balloon's height when the stone hits the ground, taking g=10.

WHY THIS WORKS

A released object keeps the balloon velocity it had at release. After release, the stone and balloon follow different accelerations and must be tracked with the same time interval.

METHOD

Use upward positive. Stone: $y=75+10t-5t^2$. Setting y=0 gives t=5 s. During those 5 s, the balloon rises another 50 m. Its height is 75+50=125 m.

ANSWER / CHECK

Its height is 75+50=125 m

CONCEPT TO KEEP

Keep: an object released from a moving carrier inherits the carrier velocity at the release instant.

EX-M9C-01

Set 6 - Q2

QUESTION RECAP

From a building, Ball A is thrown upward and Ball B downward with the same speed magnitude. Compare their velocities on reaching the ground, neglecting air resistance.

WHY THIS WORKS

With no air resistance, final speed at a given height depends on initial speed magnitude and height change, not on whether the initial direction was up or down.

METHOD

Both balls begin with the same speed magnitude u at the same height H above ground. For either ball, the final speed satisfies $v^2=u^2+2gH$. Therefore the impact speed magnitudes are equal. Since both are moving downward at impact, their velocity vectors are also equal in this one-dimensional setup: $v_A=v_B$.

ANSWER / CHECK

Since both are moving downward at impact, their velocity vectors are also equal in this one-dimensional setup: $v_A=v_B$

CONCEPT TO KEEP

Keep: without air resistance, equal initial speed magnitudes from the same height give equal impact speed magnitudes.

CHECK YOUR METHOD - SET 6 / 8

Paired launches & first graph reading

Why -> method -> answer -> concept to keep.

EX-M9D-01

Set 6 - Q3

QUESTION RECAP

A body thrown upward from a tower reaches ground in t_1. Thrown downward with the same speed it reaches in t_2. Find the time if simply dropped.

WHY THIS WORKS

The upward, downward and dropped cases share the same tower height. Combining the paired launch-time relations removes the launch speed and leaves a relation for the drop time.

METHOD

The upward and downward launches use the same tower height and the same launch-speed magnitude. Combining the two constant-gravity time relations eliminates the launch speed. The source-supported paired-time result is $t_0^2 = t_1t_2$, so $t_0 = \sqrt{(t_1t_2)}$.

ANSWER / CHECK

$t_0 = \sqrt{(t_1t_2)}$

CONCEPT TO KEEP

Keep: for equal-magnitude upward/downward tower launches, the drop-from-rest time satisfies $t_0 = \sqrt{(t_1t_2)}$.

EX-M9D-02

Set 6 - Q4

QUESTION RECAP

From a tower, upward launch reaches ground in 6 s and equal-speed downward launch in 1.5 s. Find the drop-from-rest time.

WHY THIS WORKS

The upward, downward and dropped cases share the same tower height. Combining the paired launch-time relations removes the launch speed and leaves a relation for the drop time.

METHOD

Use the paired-time relation for the same tower and equal launch-speed magnitude: $t_0 = \sqrt{(t_1t_2)}$. Substituting 6 s and 1.5 s gives $t_0 = \sqrt{(9)} = 3$ s.

ANSWER / CHECK

3 s

CONCEPT TO KEEP

Keep: for equal-magnitude upward/downward tower launches, the drop-from-rest time satisfies $t_0 = \sqrt{(t_1t_2)}$.

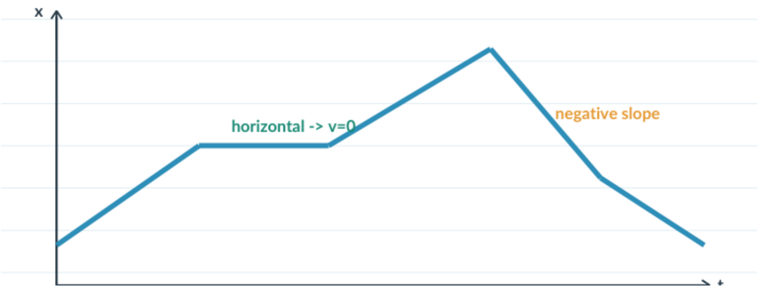
EX-M10A-01

Set 6 - Q5

QUESTION RECAP

From the supplied displacement-time graph, explain how to obtain average velocity and instantaneous velocity using secant and tangent slopes.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

On an x-t graph, velocity is encoded by slope: a secant slope gives average velocity over an interval, while a tangent slope gives instantaneous velocity.

METHOD

Use endpoint slope for average velocity and tangent slope for instantaneous velocity. The supported set on ExamSIDE is (B), (C), (E).

ANSWER / CHECK

Source-supported set: B, C and E. Use secant slope for average velocity and tangent slope for instantaneous velocity.

CONCEPT TO KEEP

Keep: x-t slope is velocity; secant = average, tangent = instantaneous.

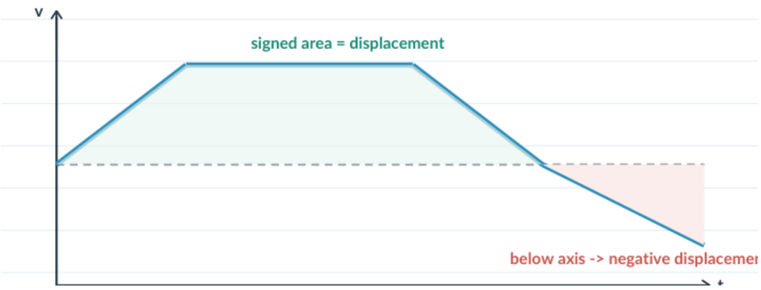
EX-M10B-01

Set 6 - Q6

QUESTION RECAP

From a v-t graph over 40 s, find total distance and average velocity.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Add absolute triangular areas for distance; add signed areas for displacement. Result: 100 m and average velocity 0.

ANSWER / CHECK

100 m and average velocity 0

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

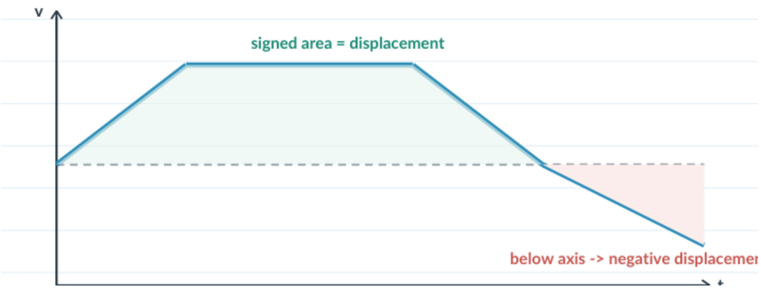
EX-M10B-02

Set 6 - Q7

QUESTION RECAP

Find distance from t=0 to 4 s from a v-t graph.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Because velocity is positive over the interval, distance equals the area under the v-t graph. Split the graph into the 0-2 s triangle and the 2-4 s rectangle: 10 m + 20 m = 30 m.

ANSWER / CHECK

30 m

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

CHECK YOUR METHOD - SET 7 / 8

Velocity-time graphs & graph families

Why -> method -> answer -> concept to keep.

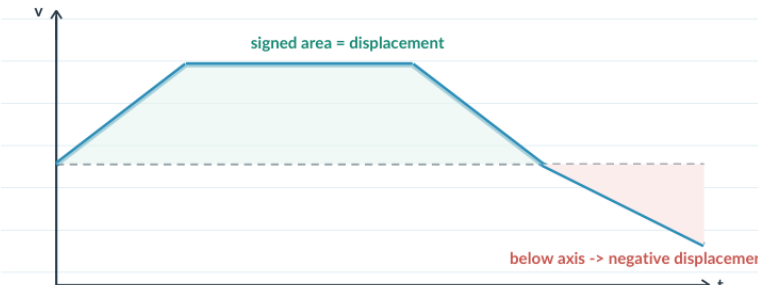
EX-M10B-03

Set 7 - Q1

QUESTION RECAP

Find airplane distance in the first 30.5 s from its v-t graph.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Distance from a v-t graph is obtained by summing the geometric areas under the curve over the requested 30.5 s. Break the graph into the source-supported simple regions, keep the graph units consistent, and add them. The published result is 12 km.

ANSWER / CHECK

12 km

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

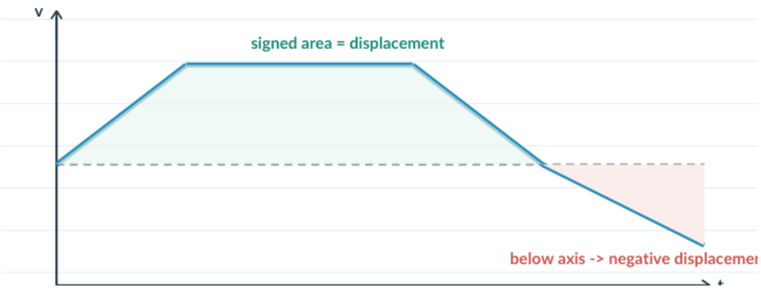
EX-M10B-04

Set 7 - Q2

QUESTION RECAP

Find displacement : distance from a v-t graph over 0-10 s.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Compute the positive and negative v-t areas separately. Signed area gives displacement = 16 m, while the sum of area magnitudes gives distance = 48 m. Hence displacement : distance = 16:48 = 1:3.

ANSWER / CHECK

1 : 3

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

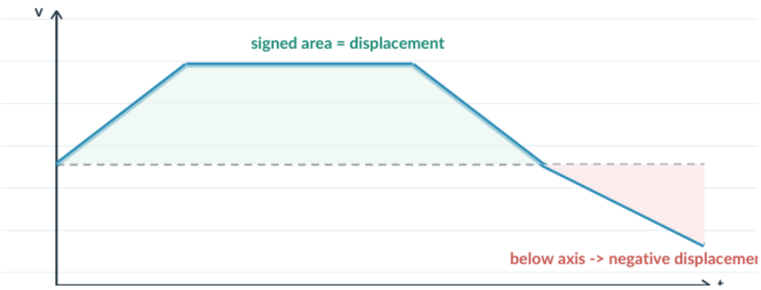
EX-M10B-05

Set 7 - Q3

QUESTION RECAP

Judge: area under v-t gives distance; area under a-t gives change in velocity.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Statement I is false in general: signed v-t area is displacement. Statement II is true. Choose 'I false, II true'.

ANSWER / CHECK

Choose 'I false, II true'

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

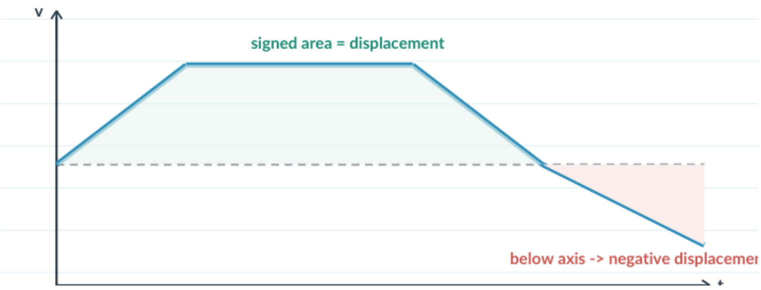
EX-M10B-06

Set 7 - Q4

QUESTION RECAP

From a v-t graph, find distance/displacement ratio in 25 s.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

A v-t graph encodes displacement as signed area. Total distance uses the magnitudes of the positive and negative areas, so sign handling is the key distinction.

METHOD

Use signed v-t area for displacement and absolute area for distance. The source graph gives distance = 250 m and displacement = 150 m, so distance/displacement = 250/150 = 5/3.

ANSWER / CHECK

5/3

CONCEPT TO KEEP

Keep: v-t signed area = displacement, absolute area sum = distance.

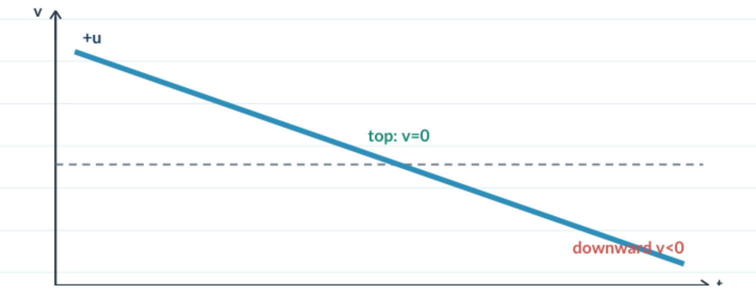
EX-M10C-01

Set 7 - Q5

QUESTION RECAP

Sketch or describe the v-t graph for a body thrown vertically upward.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

Under constant gravity, velocity changes linearly with time. A collision or inelastic stop appears as a discontinuous jump in velocity, followed by the new motion state.

METHOD

For a vertical throw, acceleration is the constant -g. Therefore v = u - gt is a straight line with constant negative slope. It crosses v = 0 at the highest point and then continues into negative velocity during the descent.

ANSWER / CHECK

A straight v-t line with constant negative slope; it crosses v=0 at the top and continues into negative velocity.

CONCEPT TO KEEP

Keep: constant acceleration gives a straight v-t segment; an instantaneous collision can create a vertical jump.

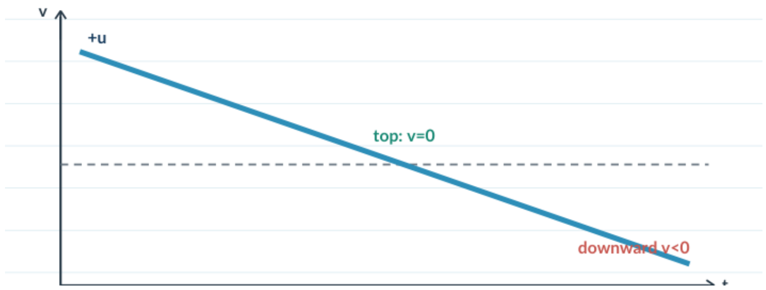
EX-M10C-02

Set 7 - Q6

QUESTION RECAP

Sketch or describe the v-t curve for a bullet moving downward for 10 s, hitting the ground inelastically, then remaining at rest.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

Under constant gravity, velocity changes linearly with time. A collision or inelastic stop appears as a discontinuous jump in velocity, followed by the new motion state.

METHOD

Before impact, v increases linearly from 100 m/s with slope +g. At 10 s the speed is 200 m/s; the inelastic impact makes v jump to 0 and then remain 0.

ANSWER / CHECK

With downward positive: v rises linearly from 100 to 200 m/s by 10 s, drops instantly to 0 at impact, then stays at 0.

CONCEPT TO KEEP

Keep: constant acceleration gives a straight v-t segment; an instantaneous collision can create a vertical jump.

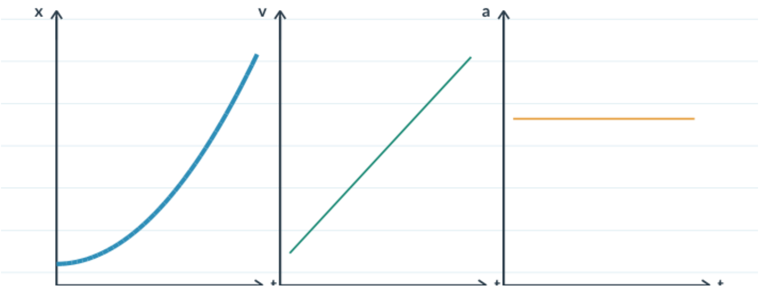
EX-M10D-01

Set 7 - Q7

QUESTION RECAP

Sketch or describe the x-t, v-t and a-t graph shapes for motion with constant acceleration.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

Constant acceleration fixes a family of graph shapes: a-t is constant, v-t is linear and x-t is quadratic. The graphs must tell one consistent motion story.

METHOD

Start from constant acceleration: the a-t graph is horizontal. Integrating the motion story qualitatively, velocity changes linearly with time and position changes quadratically. Therefore x-t is parabolic, v-t is straight and a-t is horizontal.

ANSWER / CHECK

x-t is parabolic, v-t is a straight line, and a-t is horizontal.

CONCEPT TO KEEP

Keep: constant a -> horizontal a-t, linear v-t, quadratic x-t.

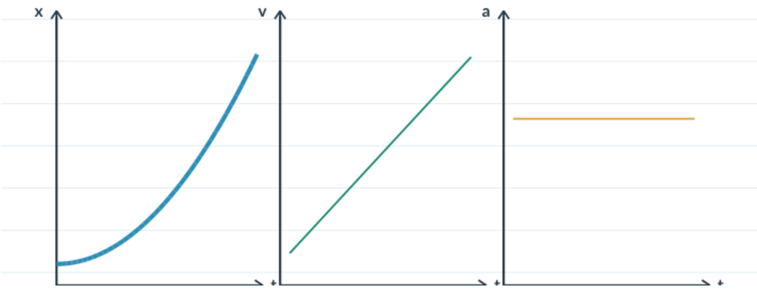
EX-M10D-02

Set 7 - Q8

QUESTION RECAP

From rest with uniform positive acceleration, identify all correct qualitative graphs.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

Constant acceleration fixes a family of graph shapes: a-t is constant, v-t is linear and x-t is quadratic. The graphs must tell one consistent motion story.

METHOD

From rest with constant positive acceleration, a is a positive constant, so a-t is horizontal above zero. Then v = at is linear through the origin and x = 1/2 at^2 is an upward-opening parabola. Compare each candidate graph with these three requirements.

ANSWER / CHECK

graphs A, B and D

CONCEPT TO KEEP

Keep: constant a -> horizontal a-t, linear v-t, quadratic x-t.

CHECK YOUR METHOD - SET 8 / 8

Advanced graph models & pattern review

Why -> method -> answer -> concept to keep.

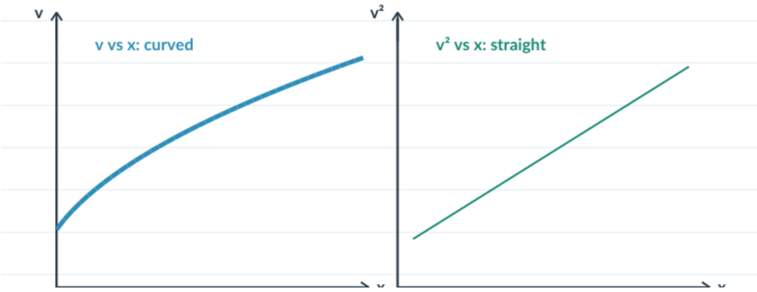
EX-M10E-01

Set 8 - Q1

QUESTION RECAP

For the given straight-line v-x graph, sketch or describe the corresponding a-x graph.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

When velocity is given as a function of position, use a = v dv/dx to translate a v-x representation into acceleration information. Shape and slope must then agree with that relation.

METHOD

Use the foundation bridge a=v dv/dx. For v=c-mx, a=-m(c-mx): a straight line with positive slope and negative intercept. ExamSIDE answer: option D.

ANSWER / CHECK

Option D - the a-x graph is a straight line with positive slope and negative intercept.

CONCEPT TO KEEP

Keep: for v(x), use a = v dv/dx; do not infer acceleration from the visual slope of v-x alone.

CHECK YOUR METHOD - SET 8 / 8

Advanced graph models & pattern review

Why -> method -> answer -> concept to keep.

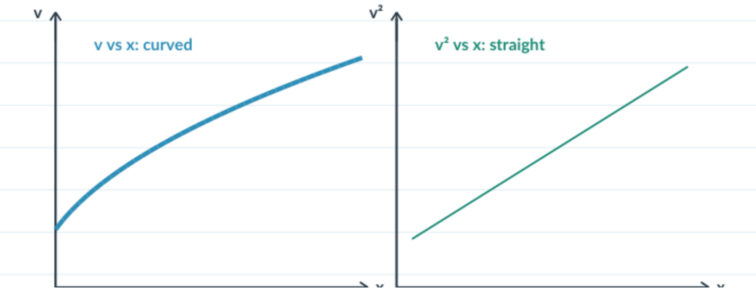
EX-M10E-02

Set 8 - Q2

QUESTION RECAP

For a linear v-x graph, sketch or describe the corresponding a-x graph.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

When velocity is given as a function of position, use a = v dv/dx to translate a v-x representation into acceleration information. Shape and slope must then agree with that relation.

METHOD

Use a=v dv/dx. Result is a straight line with positive slope and negative intercept; ExamSIDE explanation selects option A.

ANSWER / CHECK

Option A - the a-x graph is a straight line with positive slope and negative intercept.

CONCEPT TO KEEP

Keep: for v(x), use a = v dv/dx; do not infer acceleration from the visual slope of v-x alone.

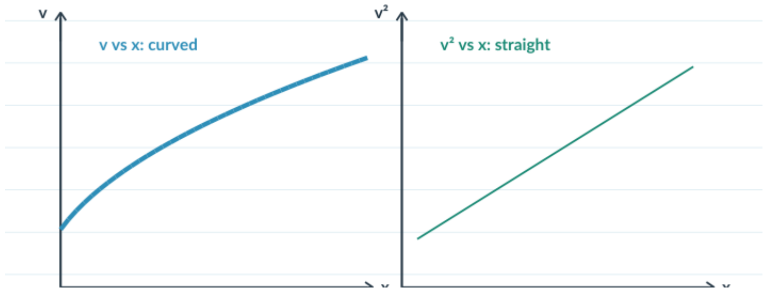
EX-M10E-03

Set 8 - Q3

QUESTION RECAP

A v^2-x graph is a straight line. Find constant acceleration from its slope.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

When velocity is given as a function of position, use a = v dv/dx to translate a v-x representation into acceleration information. Shape and slope must then agree with that relation.

METHOD

Compare $v^2=u^2+2ax$: slope of v^2 vs x equals 2a. Given slope 2, a=1 m/s^2.

ANSWER / CHECK

1 m/s^2

CONCEPT TO KEEP

Keep: for v(x), use a = v dv/dx; do not infer acceleration from the visual slope of v-x alone.

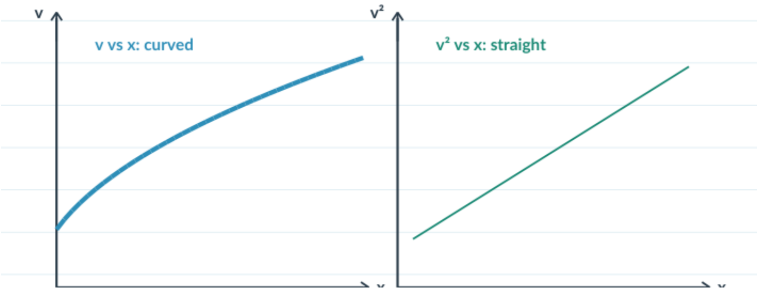
EX-M10E-04

Set 8 - Q4

QUESTION RECAP

For positive velocity with constant negative acceleration, sketch or describe the v-x graph.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

When velocity is given as a function of position, use a = v dv/dx to translate a v-x representation into acceleration information. Shape and slope must then agree with that relation.

METHOD

Using $v^2=u^2+2ax$ with $a<0$, v decreases nonlinearly with x. Correct graph is option C.

ANSWER / CHECK

Option C - positive v decreases nonlinearly with x for constant negative acceleration.

CONCEPT TO KEEP

Keep: for v(x), use a = v dv/dx; do not infer acceleration from the visual slope of v-x alone.

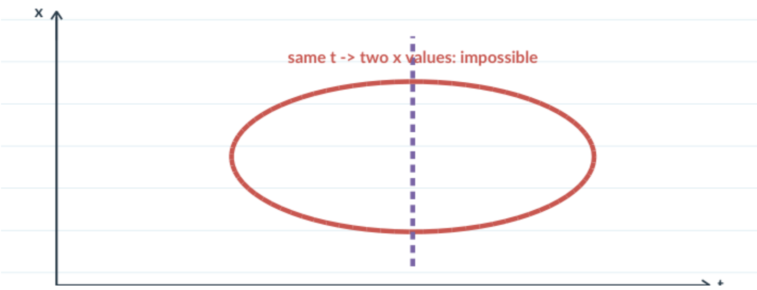
EX-M10F-01

Set 8 - Q5

QUESTION RECAP

State the single-valued-time test used to decide whether a plotted curve can represent one-dimensional motion. The original option plots are not preserved in the audited source PDF.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

Different motion graphs are valid only if they can describe one consistent one-dimensional state at each instant. Axis meaning, single-valued time and slope relations are the consistency tests.

METHOD

Apply axis meaning and single-valued-time logic. The supported answer is A, B and D only; the excluded curve violates the time/representation constraint.

ANSWER / CHECK

A, B and D only - the excluded curve violates the single-valued-time requirement.

CONCEPT TO KEEP

Keep: every graph must respect its axis meaning and describe the same state at the same time.

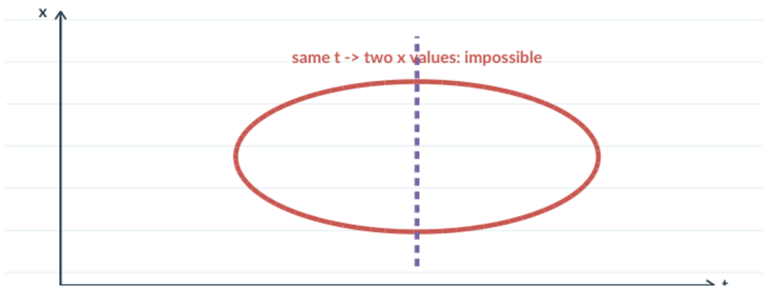
EX-M10F-02

Set 8 - Q6

QUESTION RECAP

State the consistency test for multiple graph representations of the same motion. The audited source records graph 3 as the inconsistent source option.

QUESTION FIGURE - FROM SOURCE



WHY THIS WORKS

Different motion graphs are valid only if they can describe one consistent one-dimensional state at each instant. Axis meaning, single-valued time and slope relations are the consistency tests.

METHOD

Translate each graph into the same motion story. Official-solution sources identify graph 3 as inconsistent because its initial slope implies zero initial velocity while the others do not.

ANSWER / CHECK

Graph 3 - its initial slope implies zero initial velocity while the other representations do not.

CONCEPT TO KEEP

Keep: every graph must respect its axis meaning and describe the same state at the same time.

CHECK YOUR METHOD - SET 8 / 8

Advanced graph models & pattern review

Why -> method -> answer -> concept to keep.

EX-M11D-01

Set 8 - Q7

QUESTION RECAP

For motion from rest, the ratio S_{n-1}/S_n of two consecutive one-second distances is written as 1-2/x for n=10. Find x.

WHY THIS WORKS

For motion from rest with constant acceleration, distances in successive one-second intervals follow the odd-number pattern 1:3:5:... . Ratios therefore depend only on the interval number.

METHOD

For motion from rest, the nth-second distance is proportional to 2n-1. Thus $s_9:s_{10}$ = 17:19. Set 17/19 = 1 - 2/x; then 2/x = 2/19, giving x = 19.

ANSWER / CHECK

x = 19

CONCEPT TO KEEP

Keep: from rest at constant acceleration, successive one-second distances are proportional to 1,3,5,...,2n-1.